

A Hybrid Laplacian Salp Swarm and Variable Neighbourhood Search Algorithm for Continuous Wind Farm Layout Optimization

Prince Solanki, Pulkit Dwivedi, Vanita Garg, and Vinay Shukla

Abstract—The Wind Farm Layout Optimization Problem (WFLOP) is a nonlinear constrained optimization problem in which turbine locations interact through wake-induced velocity deficits. This paper proposes LX-SSA-VNS, a two-phase hybrid for continuous WFLOP. The Laplacian Salp Swarm Algorithm (LX-SSA), previously introduced by Solanki and Deep [30], explores the layout space during the first half of the evaluation budget. Variable neighbourhood search (VNS) then intensifies the best layout found, using nested turbine-perturbation neighbourhoods and an adaptive local search. Both phases share a penalized objective with Jensen wake interactions, direction-dependent Weibull wind statistics and explicit farm-boundary and minimum-spacing constraints. The hybrid is compared with LX-SSA, SSA, particle swarm optimization (PSO), differential evolution (DE), the original basic VNS (BVNS), a modified standalone VNS and a gradient-based multistart SLSQP solver on 68 benchmark cases (two wind data sets, three circular farms, 2–15 turbines). Each comparison uses 30 seed-paired runs at an equal budget of 6,030 objective evaluations and is analyzed with Friedman, Holm-corrected Wilcoxon signed-rank and effect-size statistics, together with feasibility rates, convergence curves and an ablation of the budget split. LX-SSA-VNS attains the second-best average rank (2.74) and a feasibility rate of 98.1%. It is significantly better than LX-SSA in 35 cases and never worse, and it ranks significantly better than SSA, PSO, DE, LX-SSA and SLSQP. The modified VNS remains the best method (average rank 2.09): it is statistically equivalent to the hybrid in 53 cases and better in 15 dense cases, and the ablation attributes the gain of the hybrid to its VNS phase. The original BVNS ranks third (3.62) and is the only method feasible in every run; the modified VNS outperforms it for layouts with six or more turbines. On the Horns Rev 1 offshore site a similar ordering appears, no method improves on the installed layout within the budget, and only the constraint-aware SLSQP finds feasible layouts for the full 80-turbine farm. The algorithm ordering is insensitive to the power-curve model but sensitive to the wake model, and absolute energy values are strongly model-dependent.

Index Terms—Hybrid metaheuristic, Laplacian Salp Swarm Algorithm (LX-SSA), Variable neighbourhood search, Wake effect, Wind farm layout optimization.

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I. INTRODUCTION

Wind energy has emerged as a key component of sustainable energy systems. The efficient design of wind farms is governed by the Wind Farm Layout Optimization Problem (WFLOP), which aims to maximize energy production while minimizing wake-induced losses under spatial constraints. The nonlinear and highly coupled nature of wake interactions makes WFLOP a challenging optimization problem.

A wide range of optimization techniques have been applied to WFLOP, including Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), Ant Colony Optimization (ACO), and hybrid approaches. While these methods have shown effectiveness, many suffer from issues such as premature convergence, sensitivity to parameter settings, and limited exploration capability in high-dimensional continuous search spaces.

Recently, swarm intelligence methods such as the Salp Swarm Algorithm (SSA) have gained attention due to their simplicity and low computational complexity. However, standard SSA often exhibits limited exploration capability and may become trapped in local optima, particularly in complex continuous optimization problems such as WFLOP.

Solanki and Deep [30] introduced the Laplacian Salp Swarm Algorithm (LX-SSA) in 2023 as an enhanced variant of SSA in which a Laplace-distribution-based position update is used to improve exploration and population diversity. The design, motivation and benchmark-function validation of LX-SSA belong to that earlier publication. Like other population-based methods, LX-SSA spends many evaluations on small improvements once its swarm has contracted around the food source. Trajectory-based methods such as variable neighbourhood search (VNS) [40] have the opposite profile: they refine a single incumbent efficiently but depend on a good starting point. The present paper combines the two in a hybrid, LX-SSA-VNS, and evaluates it on continuous WFLOP, where each candidate solution represents turbine coordinates and must satisfy wake-dependent and geometric feasibility requirements.

The standard WFLOP ingredients—Jensen’s wake model, a Weibull wind representation, and a benchmark turbine power model—are retained deliberately to permit comparison with the established WFLOP literature.

The contributions of this paper are summarized as follows:

- a two-phase hybrid, LX-SSA-VNS, in which the previously published LX-SSA [30] explores the continuous

layout space and a VNS with nested turbine-perturbation neighbourhoods and an adaptive local search intensifies the best layout (Section VI);

- a continuous turbine-coordinate formulation with explicit farm-boundary and minimum-spacing constraints, shared by both phases, within the standard Jensen–Weibull WFLOP benchmark framework;
- a controlled comparison of the hybrid with LX-SSA, SSA, PSO, DE, the original basic VNS, a modified stand-alone VNS and a gradient-based multistart SLSQP solver on 68 benchmark cases at equal objective-evaluation budgets, with 30 seed-paired runs per method and case, full statistical analysis (Friedman and Holm-corrected Wilcoxon signed-rank tests, effect sizes), feasibility rates and convergence curves (Section VIII);
- a comparison of the original and the modified VNS (Section VIII-F) and an ablation that separates the contributions of the two phases and examines the budget split (Section VIII-G);
- an application to the Horns Rev 1 offshore site with its real turbine, wind climate and outline, compared against the installed layout (Section VIII-J); and
- a sensitivity analysis of the benchmark assumptions (power curve, wake model, minimum spacing and boundary handling; Section IX).

Renewable energy systems are increasingly being used as alternatives to conventional fossil fuel resources and climate change since they can decrease greenhouse gas (GHG) emissions, transmission losses, and transformation losses. Wind energy is a significant contributor to sustainable energy resources. Wind energy is typically harnessed through wind turbines, which convert the kinetic energy of moving air into mechanical energy and subsequently into electricity.

The efficient utilization of this resource leads to the Wind Farm Layout Optimization Problem (WFLOP), which involves determining the optimal spatial arrangement of wind turbines within a defined area to maximize energy production while considering constraints such as wake effect, turbine spacing, etc. The wake effect is a critical factor that must be carefully considered in the design of a wind farm layout. The wake produced by an upstream turbine leads to a reduction in wind speed, thus reducing the performance of downstream turbines. The wake effect can be modeled using generalized wake effect models that incorporate thrust and associated turbulence level. The wake spreads out downwind and its velocity deficit decays with distance. Packing more turbines into a fixed area raises installed capacity but also brings turbines closer together, which increases wake interaction and reduces the energy produced per turbine; minimum-spacing constraints are therefore imposed, and their value directly affects how many turbines can be placed (see Section IX).

Nowadays, the production of wind energy is increasing rapidly, and it is likely to increase further in the coming years. The cost associated with generating wind energy poses a major challenge during the initial stage of wind farm development. One of the most effective strategies to mitigate these costs is through the optimization of the wind farm layout, which can significantly enhance energy capture while

minimizing infrastructure and operational expenses. Over the years, researchers have used various optimization techniques and algorithms, including Genetic Algorithm (GA), Simulated Annealing (SA), Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and Biogeography-Based Optimization (BBO) to address this problem.

Mosetti et al. [1] and Grady et al. [2] solved WFLOP by using a binary-coded GA to obtain the maximum energy output with the minimum installation costs. Ozturk and Norman [3] continued Mosetti's work on WFLOP with a different objective function and a heuristic approach that considered a continuous representation instead of a discrete one. The authors proposed a methodology for optimizing wind farm layouts with the objective of maximizing annual energy production (AEP), while considering key environmental and operational factors such as wind speed, wind direction, and wake interactions. Lackner and Elkinton [4] proposed an analytical model for optimizing offshore wind farms. The model's objective function was designed to maximize the annual power output of the wind farm, which is inherently dependent on turbine placement and wake effects. Castro et al. [5] used GA to design the optimal layout of the wind farms. The objective function aimed to maximize the farm's annual energy output, subject to constraints such as turbine spacing, wind speed, etc. Huang [6] used a distributed GA to find the optimal layout of wind turbines in large wind farms to obtain the maximum annual profit, which depends on factors such as energy output, capital cost, and operating cost. Elkinton et al. [7] employed a methodology for offshore wind farm layout optimization that used various optimization algorithms. To evaluate the effectiveness of their approach, the authors compared the performance of four well-known optimization algorithms: Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Simulated Annealing (SA). The objective of the optimization problem was to obtain the maximum energy output subject to some constraints. Bilbao and Alba [8] used SA to optimize the locations of wind turbines to maximize annual profit (as in Huang [6]). Rivas et al. [9] also used SA that performs three types of local search operations iteratively to find the optimal layout of the turbines in a large offshore wind farm. Emami and Nogreh [10] proposed a novel objective function that includes the construction cost of a wind farm to solve WFLOP using GA. The overarching goal was to maximize the annual energy yield while simultaneously minimizing the total construction cost. Sisbot et al. [11] used a multi-objective GA to optimize wind farm layout design. The algorithm maximized energy production while minimizing the installation cost of turbines. Kusiak and Song [12] developed a wind distribution-based turbine placement model that considered wake loss as a function of wind turbine locations and wind direction. This problem was reformulated into a bi-criteria optimization task aimed at simultaneously maximizing energy output and minimizing constraint violations. To address this, the authors employed a multi-objective Evolutionary Strategy (ES) to derive optimal solutions. Eroglu et al. [13], [14] proposed Ant Colony Optimization (ACO) and Particle Filtering (PF) approaches to find an optimal solution by iteratively improving the layout

of the wind turbines to maximize the expected energy output while maintaining wake loss. Samorani [15] demonstrated the trade-off between maximizing power output and minimizing the wake effect within several turbines while solving WFLOP. Bansal and Farswan [16] employed Biogeography-based Optimization (BBO) to optimize the wind farm layout, proposing a maximum count of turbines that should be placed in a wind farm to obtain maximum power with minimum wake loss. Ant Colony Optimization (ACO) has been employed to optimize the collector system topology of offshore wind farms by formulating the design problem as a multiple traveling salesman problem, resulting in reduced cable length and associated costs [27]. Ju and Liu [17] proposed two novel algorithms called Adaptive Genetic Algorithm (AGA) and Self-Informed Genetic Algorithm (SIGA) that incorporate the self-adaptive capability of individuals in a Conventional Genetic Algorithm (CGA) to solve WFLOP. Jin *et al.* [18] developed a power loss cost model for offshore wind farms that accounts for the wake effect on the wind turbine output and used an Adaptive Particle Swarm Optimization (APSO) algorithm to optimize the layout design. A comprehensive review of state of the art advances in WFLO and electrical system design, including cable connection scheme optimization, is provided in [28]. Dhiman *et al.* [19] proposed a wake management technique that uses lidar (light detection and ranging) simulations to detect the wake's location and intensity and then redirects the downstream turbines' yaw angle to reduce the wake effect. Nagpal *et al.* [20] proposed a local, continuous refinement approach to refine the layouts generated by a gradient-free heuristic algorithm called Distributive Genetic Algorithm (DGA). Particle Swarm Optimization (PSO) has been successfully employed for solving WFLO to reduce wake effects and improve overall power output [29]. Bai *et al.* [21] proposed an enhanced Adaptive Genetic Algorithm (AGA) for solving the optimal layout problem of wind turbines in a wind farm. The authors modeled the layout optimization task as a single-player reinforcement learning problem and integrated Monte-Carlo tree search into an evolutionary algorithm to enhance its exploitation capability. The proposed algorithm was then applied to optimize the layout of a recently approved wind farm project in New Jersey. Cazzaro and Pisinger [22] developed an advanced heuristic for solving WFLOP using the Variable Neighborhood Search (VNS) meta-heuristic. The heuristic considers the turbine spacing constraint and construction costs in the optimization process. The WFLO literature has also moved beyond classical population-based metaheuristics. Pseudo-gradient methods have been developed to obtain faster layout refinement for large design spaces [33]; three-dimensional Gaussian wake formulations have been coupled with layout optimization to improve wake representation [34]; machine-learning wake surrogates have been used for large-scale layout design [35]; CFD-based Kriging surrogates have been introduced to reduce the computational burden of higher-fidelity evaluations [36]; and recent work combines data-driven and physics-informed models for large-scale WFLO [37], [38]. These developments show that no single optimizer or wake model is universally dominant: current research balances wake-model fidelity, computational cost, scalability, and opti-

mization robustness.

Against this background, the purpose of the present study is deliberately narrower than proposing a new WFLO paradigm. LX-SSA was already introduced and benchmarked as a continuous optimizer in 2023 [30]. Here it is combined with VNS, and the hybrid is examined in a standard continuous WFLO benchmark under geometric constraints and wake-coupled objectives. Jensen's model and the benchmark Weibull/power formulation are retained primarily for comparability with the established literature; they are not presented as state-of-the-art wake physics or as novel modeling contributions.

II. MOTIVATION FOR THE HYBRID DESIGN

Continuous WFLOP contains $2N$ spatial decision variables for N turbines, nonlinear wake coupling, and geometric feasibility constraints. Such features can produce many locally attractive layouts and make diversity preservation relevant to a population-based search.

In the 2023 LX-SSA, the standard SSA follower dynamics are supplemented by a Laplace-distribution-based candidate update and greedy retention [30], [31]. The heavy-tailed Laplace perturbation can generate both short and occasional longer moves, which provides a plausible mechanism for exploring spatially separated layout configurations. In a swarm, however, these moves are spread over the whole population, so the final refinement of the best layout, which in WFLOP means small coordinated moves of individual turbines along the spacing constraints, proceeds slowly.

VNS [40], which has been applied successfully to discrete WFLOP [22], addresses exactly this refinement: it perturbs one, two or three turbines at a time with increasing step sizes and keeps only improvements. Its weakness is its dependence on the starting layout. A sequential hybrid, in which LX-SSA supplies the starting layout for VNS, is therefore a natural combination. Whether it pays off at a fixed budget, compared with running either component alone, is an empirical question: the experiments answer it by comparing the hybrid with LX-SSA, with stand-alone VNS in its original and modified forms and with four further baselines at equal objective-evaluation budgets (Section VIII). Pseudo-gradient, surrogate and data-driven WFLO methods remain literature context.

The remainder of the paper presents the benchmark model, summarizes the previously published LX-SSA to the extent needed for reproducibility, defines its WFLOP mapping and constraint handling, introduces the hybrid, reports the experimental results, and finally states the limitations of the evidence.

III. PROBLEM MODELING

Certain definitions and presumptions are necessary to design a wind farm and determine the optimal placements of turbines, which are as follows:

- 1) **Installed Capacity and Prescribed Turbine Count:** For identical turbines, nameplate capacity scales linearly with the number of installed units, but the *expected farm power does not* because wake interactions depend on the layout. In each optimization run considered here, the

turbine count N is prescribed in advance and LX-SSA optimizes only the $2N$ planar coordinates. Results across several values of N therefore constitute a turbine-count sweep, not simultaneous optimization of N and turbine locations.

- 2) **Turbine Location Representation:** The location of each turbine is represented in 2-dimensional space (ρ, σ) , and the magnitude of its location vector is computed as $\sqrt{\rho^2 + \sigma^2}$.
- 3) **Uniformity of Turbines:** All turbines should be uniform in terms of internal quality (power curve, capacity) and external quality (design, model, hub height).
- 4) **Wind Speed Modeling:** To characterize wind speed variability, the Weibull distribution is commonly used to model its probability density function (PDF). For a specific location, height, and wind direction, the wind speed s is assumed to follow a Weibull distribution:

$$p_s(s; \zeta, \psi) = \frac{\zeta}{\psi} \left(\frac{s}{\psi} \right)^{\zeta-1} \exp \left(- \left(\frac{s}{\psi} \right)^{\zeta} \right),$$

where ζ corresponds to the shape parameter, ψ is the scale parameter, and $p_s(\cdot)$ represents the PDF.

- 5) **Directional Functions:** Wind speed (s) is dependent on the wind direction θ , expressed as $s = s(\theta)$. Similarly, both the shape parameter (ζ) and the scale parameter (ψ) vary as functions of wind direction, i.e., $\zeta = \zeta(\theta)$, $\psi = \psi(\theta)$, $0^\circ \leq \theta \leq 360^\circ$. Therefore, θ is an important parameter in WFLOP. Figure 1 illustrates a wind farm with four representative wind directions, with $\theta = 0^\circ$ representing the east, $\theta = 90^\circ$ representing the north, $\theta = 180^\circ$ representing the west, and $\theta = 270^\circ$ representing the south.
- 6) **Minimum Turbine Spacing:** To preserve comparability with the legacy benchmark studies used in the numerical section, the minimum center-to-center spacing is set to $4D = 8R$, where $D = 2R$ is the rotor diameter:

$$(\rho_i - \rho_j)^2 + (\sigma_i - \sigma_j)^2 \geq 64R^2.$$

This value is a *benchmark lower bound* retained for comparison with the legacy WFLOP literature [12], [16], not a universal engineering recommendation. Project-specific spacing requirements can be larger and direction-dependent; therefore conclusions about feasible turbine density are conditional on the present $4D$ assumption. Section IX-B quantifies this dependence with a $4D/5D/6D$ sensitivity study of geometric capacity, attained objective and wake loss.

- 7) **Wind Farm Shape:** WFLOP focuses on determining the optimal spatial placement of wind turbines within a specified area. The shape of the wind farm can have an impact on the optimal layout. In WFLOP, the shape can be circular, rectangular, or any other feasible shape. For this study, a circular wind farm is chosen for consistency with the legacy benchmark studies [12], [16]; a circular domain also has no preferred orientation, so the boundary does not favor any wind direction. No claim is made that a circular site is superior in land use to other shapes.

- 8) **Farm Boundary Constraint:** To ensure that all the wind turbines are placed within the designated wind farm area, the location of each turbine $T_i(\rho_i, \sigma_i)$ must adhere to the following spatial constraint:

$$\rho_i^2 + \sigma_i^2 \leq r^2,$$

where r is the wind farm radius. This study examines three cases with radii of 500 m, 750 m, and 1000 m.

- 9) **Search Space:** In WFLOP, the search space is defined by the boundary of the wind farm, which is circular in shape as shown in Figure 1. Since the search space is continuous, there is no need to use a grid system to locate a wind turbine.

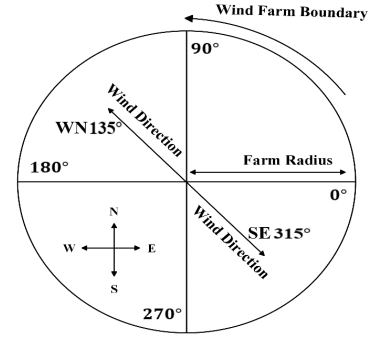


Fig. 1: A circular wind farm

- 10) **WFLOP Mathematical Model:** The mathematical model of WFLOP contains two parts: the wake effect model and the power output model. The wake effect model captures the reduction in power generation experienced by downstream turbines due to the wake induced by upstream turbines. Jensen's wake model [23], [24] is a widely used model for solving WFLOP. The power output model used in WFLOP is considered from Kusiak and Song [12].
- 11) **Objective:** This study aims to maximize the power generation by minimizing wake-induced losses, while satisfying two constraints obtained from assumptions (6) and (8).

A. The Wake Effect Model

The wake effect can lead to a substantial reduction in the wind speed experienced by downstream turbines as a result of momentum extraction and wake expansion. Therefore, accounting for wake losses is a critical aspect of wind farm layout design. Jensen's wake model [23], [24] is retained here because it is computationally inexpensive and has been used extensively in the legacy WFLOP benchmark literature against which the present results are compared. This choice is made for *benchmark comparability*, not because Jensen is the most accurate contemporary wake model. Gaussian and three-dimensional wake formulations can represent the velocity field more realistically and have been used successfully in WFLO studies [34], [39]; consequently, the numerical conclusions below are model-dependent.

When a uniform wind comes into contact with a turbine, it creates turbulence, and a wake expands linearly downstream, causing a reduction in wind speed, i.e., from upstream wind speed s_{up} to downstream wind speed s_{down} . Figure 2 illustrates the wake model of WFLOP.

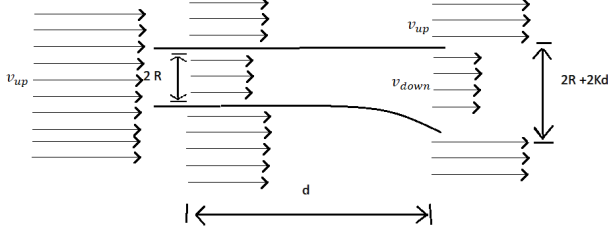


Fig. 2: Wake model representation for wind turbine layout

Let s_{up} represent the wind speed before encountering the turbine, s_{down} represent the downstream wind speed, K be the wake spreading constant, d be the inter-turbine distance, and R be the rotor radius of the turbine.

The velocity deficit induced at turbine j resulting from the wake generated by turbine i is given by:

$$\delta s_{ij} = 1 - (s_{down}/s_{up}) = \frac{1 - \sqrt{1 - C_T}}{\left(1 + \frac{K d_{ij}}{R}\right)^2}$$

where C_T denotes the turbine thrust coefficient, and d_{ij} represents the inter-turbine distance between turbines i and j .

If a turbine is influenced by the wakes of multiple upstream turbines, the total velocity deficit for that turbine is the superposition of the velocity deficits due to each wake. Assuming turbine i is influenced by the wakes of N turbines, then the overall deficit can be computed as follows:

$$\delta s_i = \sqrt{\sum_{j=1, j \neq i}^N (\delta s_{ij})^2}. \quad (1)$$

Each turbine rotor is positioned perpendicular to the prevailing wind direction θ to ensure maximum energy capture. When wind passes downstream of a turbine located at (ρ, σ) , the resulting wake can be idealized as an imaginary half cone. Figure 3 indicates this half cone generated by a turbine positioned at (ρ, σ) .

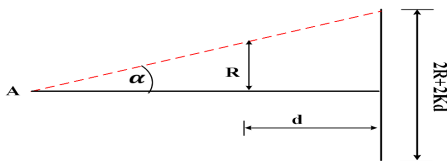


Fig. 3: Geometric representation of the wake expansion angle α

where A is the imaginary vertex of the half cone representing the wake region. The geometry of the cone is governed by the wake spreading constant K , α ($0 \leq \alpha \leq \pi/2$) is the half-cone

angle, i.e., the angle between the central axis and the side of the cone. The value of α is computed as $\alpha = \arctan(K)$. This imaginary half cone is used to model the influence of wake effects from one turbine on others within the wind farm.

Lemma 1. For a specified wind direction θ , the angle β_{ij} , where $0 \leq \beta_{ij} \leq \pi$, represents the angle between the two vectors extending from point A to turbines i and j and it can be computed using the following expression:

$$\beta_{ij} = \cos^{-1} \left(\frac{u_{ij}}{\sqrt{u_{ij}^2 + v_{ij}^2}} \right),$$

with

$$\begin{aligned} u_{ij} &= (\rho_i - \rho_j) \cos \theta + (\sigma_i - \sigma_j) \sin \theta + \frac{R}{K}, \\ v_{ij} &= -(\rho_i - \rho_j) \sin \theta + (\sigma_i - \sigma_j) \cos \theta, \end{aligned}$$

where $\frac{R}{K}$ denotes the distance from the rotor center to the virtual vertex A of the wake cone.

Lemma 2. If turbine i is positioned within the wake generated by turbine j , the distance between these, along the wind direction θ is given by:

$$d_{ij} = |(\rho_i - \rho_j) \cos \theta + (\sigma_i - \sigma_j) \sin \theta|.$$

The equation (1) is then expressed by restricting the summation to those turbines j where $j \neq i$ and $\beta_{ij} < \alpha$:

$$\delta s_i = \sqrt{\sum_{j=1, j \neq i, \beta_{ij} < \alpha}^N (\delta s_{ij})^2}. \quad (2)$$

Equation (2) indicates that δs_i is a function of both the wind direction θ and the turbine i 's location (i.e., (ρ_i, σ_i)).

The transformation of the Weibull parameter is made explicit as follows. For a fixed directional bin θ , let the free-stream speed be

$$S_\theta \sim \text{Weibull}(\zeta(\theta), \psi(\theta)).$$

Under the present Jensen benchmark, the combined deficit $\delta s_i(\theta)$ is treated as a deterministic multiplicative reduction within that directional bin, so the local inflow at turbine i is

$$S_{i,\theta} = [1 - \delta s_i(\theta)] S_\theta.$$

For a positive constant a , if $S \sim \text{Weibull}(\zeta, \psi)$, then $aS \sim \text{Weibull}(\zeta, a\psi)$. Hence the shape parameter is unchanged while the scale parameter becomes

$$\psi_i(\theta) = \psi(\theta) [1 - \delta s_i(\theta)], \quad i = 1, 2, \dots, N. \quad (3)$$

This scaling is physically consistent with the *specific simplifying assumption* that the wake model acts as a deterministic speed multiplier for a given directional bin. It should not be interpreted as a general turbulence model; more detailed wake/turbulence models may alter the full wind-speed distribution rather than only its scale.

B. The Power Model

The power model follows the simplified Kusiak–Song benchmark used by the comparison studies [12]. It is a computational surrogate for a manufacturer power curve rather than a claim of high-fidelity turbine aerodynamics:

$$f(s) = \begin{cases} 0, & s < s_{\text{cut-in}}, \\ \lambda' s + \eta, & s_{\text{cut-in}} \leq s \leq s_{\text{rated}}, \\ P_{\text{rated}}, & s > s_{\text{rated}}. \end{cases} \quad (4)$$

Here $s_{\text{cut-in}}$ and s_{rated} are the benchmark cut-in and rated speeds, while λ' and η define the linear segment. The legacy benchmark does not explicitly model a cut-out shutdown in the subsequent analytical integral; the present manuscript therefore uses the model consistently as written above. This simplification improves comparability with prior WFLOP results but can introduce error relative to a turbine-specific nonlinear power curve. Rather than assuming this error is negligible, Section IX-A re-evaluates the stored layouts with a nonlinear (cubic) power curve, with and without a 25 m/s cut-out, and reports the resulting change in expected power and in the algorithm ordering.

For a turbine positioned at (ρ, σ) and wind direction θ , the expected power output is given by:

$$\begin{aligned} E(P_i) &= \int_0^{360} P_\theta(\theta) \cdot E(P_i, \theta) d\theta \\ &= \int_0^{360} P_\theta(\theta) \cdot \int_0^\infty f(s) \cdot p_s(s, \zeta(\theta), \psi_i(\theta)) ds d\theta. \end{aligned} \quad (5)$$

Thus, the optimization problem is formulated with the objective of maximizing the total power output of the wind farm, subject to the spatial and operational constraints defined by assumptions (6) and (8). The mathematical formulation is:

$$\begin{aligned} &\text{Maximize} \quad \sum_{i=1}^N E(P_i) \\ &\text{subject to} \quad (\rho_i - \rho_j)^2 + (\sigma_i - \sigma_j)^2 \geq 64R^2, \\ &\quad \rho_i^2 + \sigma_i^2 \leq r^2, \quad i, j = 1, 2, \dots, N, \quad i \neq j \end{aligned}$$

where $E(P_i)$ corresponds to the expected power output of the i^{th} turbine under the adopted benchmark model. The piecewise-linear representation in (4) is retained for consistency with the benchmark literature, not because its approximation error is assumed negligible. Incorporating Equation (4) and the Weibull model, Equation (5) becomes:

$$\begin{aligned} E(P_i) &= \int_0^{360} p_\theta(\theta) \int_0^\infty f(s) p_s(s, \zeta(\theta), \psi_i(\theta)) ds d\theta \\ &= \int_0^{360} p_\theta(\theta) \int_0^\infty \frac{f(s) \zeta(\theta)}{\psi_i(\theta)} \left[\frac{s}{\psi_i(\theta)} \right]^{\zeta(\theta)-1} e^{-\left[\frac{s}{\psi_i(\theta)} \right]^{\zeta(\theta)}} ds d\theta \\ &= \int_0^{360} p_\theta(\theta) \left[\lambda' \int_{s_{\text{cut-in}}}^{s_{\text{rated}}} \frac{s \zeta(\theta)}{\psi_i(\theta)} \left[\frac{s}{\psi_i(\theta)} \right]^{\zeta(\theta)-1} e^{-\left[\frac{s}{\psi_i(\theta)} \right]^{\zeta(\theta)}} ds \right. \\ &\quad \left. + \eta \int_{s_{\text{cut-in}}}^{s_{\text{rated}}} \frac{\zeta(\theta)}{\psi_i(\theta)} \left(\frac{s}{\psi_i(\theta)} \right)^{\zeta(\theta)-1} e^{-\left(\frac{s}{\psi_i(\theta)} \right)^{\zeta(\theta)}} ds \right. \\ &\quad \left. + P_{\text{rated}} \int_{s_{\text{rated}}}^\infty \frac{\zeta(\theta)}{\psi_i(\theta)} \left(\frac{s}{\psi_i(\theta)} \right)^{\zeta(\theta)-1} e^{-\left(\frac{s}{\psi_i(\theta)} \right)^{\zeta(\theta)}} ds \right]. \end{aligned}$$

Wind direction and speed are discretized into $N_\theta + 1$ and $N_s + 1$ bins, respectively, using equal bin widths to approximate the integration part using the Riemann sum [25]. The discretization approach for estimating expected power output was described by Kusiak and Song [12].

Let wind direction bins be defined by $\theta_0 = 0^\circ, \theta_1, \dots, \theta_{N_\theta+1} = 360^\circ$ and wind speed bins between $s_{\text{cut-in}}$ and s_{rated} be $s_0 = s_{\text{cut-in}}, s_1, \dots, s_{N_s+1} = s_{\text{rated}}$.

Using the Riemann sum approximation [25], the expected power output of the i^{th} turbine is estimated as:

$$\begin{aligned} E(P_i) &= \lambda' \sum_{j=1}^{N_s+1} \left(\frac{s_{j-1} + s_j}{2} \right) \sum_{l=1}^{N_\theta+1} (\theta_l - \theta_{l-1}) \omega_{l-1} \\ &\quad \left[e^{-\left(\frac{s_{j-1}}{\psi_i(\frac{\theta_l + \theta_{l-1}}{2})} \right)^{\zeta(\frac{\theta_l + \theta_{l-1}}{2})}} - e^{-\left(\frac{s_j}{\psi_i(\frac{\theta_l + \theta_{l-1}}{2})} \right)^{\zeta(\frac{\theta_l + \theta_{l-1}}{2})}} \right] \\ &\quad + P_{\text{rated}} \sum_{l=1}^{N_\theta+1} (\theta_l - \theta_{l-1}) \omega_{l-1} e^{-\left(\frac{s_{\text{rated}}}{\psi_i(\frac{\theta_l + \theta_{l-1}}{2})} \right)^{\zeta(\frac{\theta_l + \theta_{l-1}}{2})}} \\ &\quad + \eta \sum_{l=1}^{N_\theta+1} (\theta_l - \theta_{l-1}) \omega_{l-1} \\ &\quad \left[e^{-\left(\frac{s_{\text{cut-in}}}{\psi_i(\frac{\theta_l + \theta_{l-1}}{2})} \right)^{\zeta(\frac{\theta_l + \theta_{l-1}}{2})}} - e^{-\left(\frac{s_{\text{rated}}}{\psi_i(\frac{\theta_l + \theta_{l-1}}{2})} \right)^{\zeta(\frac{\theta_l + \theta_{l-1}}{2})}} \right]. \end{aligned}$$

Here, N_s and N_θ denote the count of intervals for wind speed and wind direction, respectively, and ω_{l-1} is the wind blowing probability for the $(l-1)$ -th wind direction bin.

a) Interpretation of the reported benchmark objective.:

The numerical tables retain the historical discretization convention of the adopted WFLOP benchmark, including the directional-bin-width factor $(\theta_l - \theta_{l-1})$. The resulting tabulated “power” values are therefore best interpreted as *benchmark objective values under that legacy convention* for comparison with prior studies, rather than as instantaneous electrical power that must be bounded by NP_{rated} . Because all 24 bins have the same width of 15° and $\sum_l \omega_{l-1} = 1$, the reported objective equals 15 times the expected farm power in kW. For example, the free-stream (ideal) objective of a single turbine is 14045.74 for Wind Data Set I, which corresponds to an expected power of 936.38 kW, i.e., a capacity factor of 0.624 and an annual energy production (AEP) of $936.38 \times 8.76 = 8202.7$ MWh. In

general, $AEP [MWh] = (\text{objective}/15) \times 8.76$. This conversion was confirmed with the independent evaluator described in Section VII-E. This distinction is important when comparing the benchmark numbers with physical nameplate capacity.

C. Constraint Handling Mechanism

The WFLOP search must satisfy the circular farm boundary and the minimum inter-turbine spacing. The implementation used for all reported runs handles both constraints with two explicit operations: a box repair of the decision variables and a large additive penalty in the minimized objective. No other repair or displacement operator is applied.

1) *Box Repair*: Every coordinate of every candidate is clipped to the bounding square of the farm after each position update,

$$\begin{aligned}\rho_i &\leftarrow \min\{\max\{\rho_i, -r\}, r\}, \\ \sigma_i &\leftarrow \min\{\max\{\sigma_i, -r\}, r\}.\end{aligned}$$

This guarantees bounded decision variables but not the circular constraint $\rho_i^2 + \sigma_i^2 \leq r^2$, which is enforced through the penalty below.

2) *Violation Measures*: The benchmark constraints are $d_{ij} = \sqrt{(\rho_i - \rho_j)^2 + (\sigma_i - \sigma_j)^2} \geq d_{\min} = 8R$ for all $i \neq j$ and $\rho_i^2 + \sigma_i^2 \leq r^2$ for all i . Their violations are measured by

$$g_{ij}^s = d_{\min} - d_{ij}, \quad g_i^b = \rho_i^2 + \sigma_i^2 - r^2, \quad (6)$$

and a layout is feasible if and only if $g_{ij}^s \leq 0$ and $g_i^b \leq 0$ for all i and j .

3) *Penalized Objective*: All optimizers minimize the penalized wake loss

$$\begin{aligned}F_p &= \left(NE_{\text{ideal}} - \sum_{i=1}^N E(P_i) \right) + \sum_{i: g_i^b > 0} (1 + \lambda g_i^b)^2 \\ &\quad + \sum_{i < j: g_{ij}^s > 0} (1 + \lambda g_{ij}^s)^2, \quad \lambda = 10^{10},\end{aligned} \quad (7)$$

where NE_{ideal} is the wake-free value of the objective, so the first term is the wake loss reported in the tables. Any violation larger than about 10^{-7} (m or m^2) makes the penalty exceed the largest attainable wake loss, so comparing F_p values implements feasibility-first selection:

- any feasible candidate is preferred to a (non-negligibly) infeasible one;
- between two feasible candidates, the one with the smaller wake loss (larger objective) is preferred;
- between two infeasible candidates, the one with the smaller sum of squared penalty terms is preferred.

Box repair is applied after every position update and F_p is used in every comparison, sort and greedy selection of Algorithms 1 and 2. The same objective, including the penalty, is used for SSA, LX-SSA, PSO and DE. Section IX-C shows that a radial projection onto the circle, used instead of box clipping, changes the attained objective values, so the boundary-handling rule is part of the experimental protocol.

IV. PREVIOUSLY DEVELOPED LAPLACIAN SALP SWARM ALGORITHM (LX-SSA)

LX-SSA was introduced by Solanki and Deep in 2023 [30]; the algorithmic novelty belongs to that publication. The present paper only summarizes the operators needed to understand and reproduce the WFLOP application. Consistent with the published SSA/LX-SSA implementation, the sorted population is divided into a leader subgroup (first half) and a follower subgroup (second half), while the best solution found so far is retained as the food source H [26], [30]. LX-SSA keeps these SSA dynamics and supplements the follower update with a Laplace-based candidate and greedy retention [30], [31].

For a leader salp $i \leq N_p/2$, the j th coordinate is updated around the food source as

$$\rho_j^i = \begin{cases} H_j + r_1((ub_j - lb_j)r_2 + lb_j), & r_3 \geq 0.5, \\ H_j - r_1((ub_j - lb_j)r_2 + lb_j), & r_3 < 0.5, \end{cases} \quad (8)$$

where $r_2, r_3 \sim U(0, 1)$ and

$$r_1 = 2 \exp \left[- \left(\frac{4l}{L} \right)^2 \right],$$

with iteration l and maximum iteration L . For follower salps $i > N_p/2$, the standard SSA position is formed from the current and preceding salps:

$$\rho_j^i = \frac{1}{2} (\rho_j^i + \rho_j^{i-1}). \quad (9)$$

LX-SSA then forms an additional follower candidate

$$y_j^i = x_j^i + \gamma(z, \phi, \chi) (H_j - x_j^i), \quad (10)$$

where x_j^i is the current follower coordinate and

$$\gamma(z, \phi, \chi) = \begin{cases} \phi + \chi \ln(2z), & z \leq 0.5, \\ \phi - \chi \ln(2(1-z)), & z > 0.5, \end{cases} \quad (11)$$

with $z \sim U(0, 1)$, which is the inverse-distribution-function sample of a Laplace variable with location ϕ and scale χ (mean ϕ , variance $2\chi^2$). The location parameter ϕ centers the Laplace variate, whereas the scale parameter $\chi > 0$ controls its dispersion and therefore the perturbation magnitude. Smaller χ concentrates moves near the center; larger χ increases the probability of longer exploratory moves. All experiments use $\phi = 0$ and $\chi = 1$, so that γ is symmetric about zero with $P(|\gamma| > 1) = e^{-1} \approx 0.37$: most follower candidates move towards H ($\gamma > 0$) or away from it ($\gamma < 0$) by a fraction of the current distance, while a heavy tail occasionally produces moves beyond H or well away from it. These values are fixed for every WFLOP case and are not tuned to WFLOP; no WFLOP-specific sensitivity analysis of ϕ and χ was carried out. Because each follower evaluates both candidates before the population is re-evaluated, one LX-SSA iteration costs $N_p/2 \times 2 + N_p = 2N_p$ objective calls, twice the N_p calls of an SSA iteration; comparisons at equal numbers of calls are therefore reported in Section VIII.

Algorithm 1 follows the published leader/follower logic and avoids the earlier ambiguity in which a follower update appeared inside the leader branch. It also avoids hard-coding a minimization inequality; “better” is determined by

the objective sense and, for WFLOP, by the feasibility rules in Section III-C.

Algorithm 1 Previously developed LX-SSA search logic [30]

```

1: Initialize and evaluate the salp population; sort it and set
  the best solution as food source  $H$ .
2: while the termination criterion is not met do
3:   Update  $r_1$ .
4:   for salp  $i = 1, \dots, N_p$  do
5:     if  $i \leq N_p/2$  then
6:       Update leader  $i$  using (8).
7:     else
8:       Form the standard follower candidate using (9).
9:       Draw  $\gamma(z, \phi, \chi)$  using (11).
10:      Form the Laplace-based follower candidate using
        (10).
11:      Evaluate both follower candidates and retain
        the better one.
12:    end if
13:  end for
14:  Enforce variable bounds, re-evaluate the population,
  sort, and update  $H$ .
15: end while
16: return the best solution  $H$ .
```

The comparison between SSA and LX-SSA in the WFLOP experiments can therefore be interpreted as a mechanism-level comparison (an ablation) of the baseline SSA dynamics against the additional Laplace-based follower candidate.

V. APPLICATION OF LX-SSA TO THE WIND FARM LAYOUT OPTIMIZATION PROBLEM (WFLOP)

The published LX-SSA [30] is employed without algorithmic redesign. For a prescribed number N of turbines, candidate j is represented by

$$\mathbf{x}_j = (\rho_j^1, \sigma_j^1, \rho_j^2, \sigma_j^2, \dots, \rho_j^N, \sigma_j^N) \in \mathbb{R}^{2N}.$$

The optimizer therefore searches only the turbine coordinates; N is fixed during a run. The objective and constraint calculations are the WFLOP model described above.

Algorithm 2 makes the implementation sequence explicit. In particular, the candidate coordinates are updated first, box repair is applied, wake deficits and Weibull-scale adjustments are recomputed, the constraint violation is measured, and all comparisons use the penalized objective F_p , which implements feasibility-first selection (Section III-C). This ordering is essential because the objective depends on the layout-dependent wake field.

No claim is made that this mapping changes the mathematical definition of LX-SSA. Its contribution is the explicit integration of the existing optimizer with the continuous WFLOP representation and constraints.

VI. HYBRID LX-SSA-VNS ALGORITHM

Population-based swarm methods such as LX-SSA spread their search over many regions of the layout space, but once the swarm has contracted around the food source, most remaining

Algorithm 2 Application of LX-SSA to WFLOP for a prescribed turbine count N

Require: Farm radius r , prescribed N , wind data, turbine/wake parameters, population size N_p , maximum iterations T , ϕ , χ

```

1: Set dimension  $d = 2N$ , bounds  $[-r, r]^d$ , and draw  $N_p$ 
  coordinate vectors uniformly in the bounds.
2: for each candidate do
3:   For each wind-direction bin, compute wake deficits
      $\delta s_i(\theta)$ .
4:   Set  $\psi_i(\theta)$  using (3) and compute  $E(P_i)$ .
5:   Compute the violations (6) and the penalized objective
      $F_p$  from (7).
6: end for
7: Sort the population by  $F_p$  and set the best candidate as
  food source  $H$ .
8: for  $t = 1$  to  $T$  do
9:   Generate the leader and follower candidates of Algo-
     rithm 1, applying box repair to each; followers keep the
     better of their two candidates by  $F_p$ .
10:  Apply box repair, evaluate  $F_p$  for the new population
    and sort it.
11:  If the best new candidate has a smaller  $F_p$  than  $H$ ,
    replace  $H$ .
12: end for
13: return  $H$  and its turbine coordinates, objective value,
    wake loss and feasibility status.
```

evaluations produce only small improvements. Trajectory-based local search behaves the other way round. Variable neighbourhood search (VNS) [40] refines one incumbent very efficiently by systematically changing the size of its perturbations, but its result depends strongly on where it starts. This complementarity motivates a two-phase hybrid, LX-SSA-VNS, in which the previously published LX-SSA [30] provides global exploration and VNS provides intensification.

A. Phase 1: Global Exploration by LX-SSA

For a budget of B objective calls and a split fraction $\rho \in (0, 1)$, LX-SSA (Algorithm 2) runs with population N_p for

$$T_1 = \text{round}\left(\frac{\rho B - N_p}{2N_p}\right)$$

iterations, because one LX-SSA iteration costs $2N_p$ calls. Its parameter r_1 decreases over these T_1 iterations, so the swarm moves from exploration to exploitation within Phase 1. The food source H at the end of Phase 1 is the starting incumbent of Phase 2.

B. Phase 2: Intensification by VNS

Starting from $x = H$, VNS uses $k_{\max} = 3$ nested neighbourhoods. In neighbourhood k , a *shaking* step moves k randomly chosen turbines by Gaussian steps with standard deviation $\sigma_k r$, where $(\sigma_1, \sigma_2, \sigma_3) = (0.05, 0.15, 0.40)$. The shaken layout y is then improved by a first-improvement *local search* of $L = 10$ single-turbine Gaussian moves. Its step

TABLE I: Wind Data Sets I and II: Direction Intervals, Weibull Parameters, and Blowing Probabilities

Int.	Start (°)	End (°)	ζ	ψ	Prob.
1	0	15	2	13	0.000
2	15	30	2	13	0.010
3	30	45	2	13	0.010
4	45	60	2	13	0.010
5	60	75	2	13	0.010
6	75	90	2	13	0.200
7	90	105	2	13	0.600
8	105	120	2	13	0.010
9	120	135	2	13	0.010
10	135	150	2	13	0.010
11	150	165	2	13	0.010
12	165	180	2	13	0.010
13	180	195	2	13	0.010
14	195	210	2	13	0.010
15	210	225	2	13	0.010
16	225	240	2	13	0.010
17	240	255	2	13	0.010
18	255	270	2	13	0.010
19	270	285	2	13	0.010
20	285	300	2	13	0.010
21	300	315	2	13	0.010
22	315	330	2	13	0.010
23	330	345	2	13	0.010
24	345	360	2	13	0.000

Int.	Start (°)	End (°)	ζ	ψ	Prob.
1	0	15	2	7.0	0.0002
2	15	30	2	5.0	0.0080
3	30	45	2	5.0	0.0227
4	45	60	2	5.0	0.0242
5	60	75	2	5.0	0.0225
6	75	90	2	4.0	0.0339
7	90	105	2	5.0	0.0423
8	105	120	2	6.0	0.0290
9	120	135	2	7.0	0.0617
10	135	150	2	7.0	0.0813
11	150	165	2	8.0	0.0994
12	165	180	2	9.5	0.1394
13	180	195	2	10.0	0.1839
14	195	210	2	8.5	0.1115
15	210	225	2	8.5	0.0765
16	225	240	2	6.5	0.0080
17	240	255	2	4.6	0.0051
18	255	270	2	2.6	0.0019
19	270	285	2	8.0	0.0012
20	285	300	2	5.0	0.0010
21	300	315	2	6.4	0.0017
22	315	330	2	5.2	0.0031
23	330	345	2	4.5	0.0097
24	345	360	2	3.9	0.0317

length is enlarged by a factor 1.5 after a successful move and reduced by a factor 0.8 after an unsuccessful one, and it is bounded between $10^{-3}r$ and $0.5r$. If the resulting layout improves on the incumbent according to F_p , it is accepted and the search returns to $k = 1$; otherwise k is increased cyclically. Phase 2 ends when the budget B is exhausted. This is the modified VNS that is also run stand-alone in Section VIII; it differs from the original basic VNS [40], [41], which shakes the whole layout and applies a complete local search, and which is evaluated separately (BVNS, Section VIII-F). Box repair and the penalized objective (7) are used exactly as in LX-SSA, so the feasibility-first rules of Section III-C apply in both phases.

C. Parameters and Cost

The hybrid uses $N_p = 30$, $\phi = 0$, $\chi = 1$ and the VNS settings above, and it has a single new parameter, the split ρ . The default $\rho = 0.5$ assigns 3,030 of the 6,030 calls to LX-SSA ($T_1 = 50$) and the remaining 3,000 to VNS; its sensitivity is examined in Section VIII-G. The per-call cost of both phases is dominated by the objective evaluation, so the hybrid has the same $\mathcal{O}(B N^2 N_\theta N_s)$ cost as the other methods for a budget of B calls. Because the random-number generator is seeded once and Phase 1 draws the initial population first, for a given seed the hybrid starts from the same initial population as all other methods, which keeps the comparisons paired.

VII. EXPERIMENTAL SETUP

A. Test Problems

Two wind data sets are used (Table I). *Wind Data Set I* has fixed Weibull parameters ($\zeta = 2$, $\psi = 13$) in all directions and a strongly directional blowing probability: $\omega_0 = \omega_{23} = 0$, $\omega_5 = 0.2$, $\omega_6 = 0.6$, and $\omega_l = 0.01$ for all other intervals. *Wind Data Set II* keeps $\zeta = 2$ but lets the scale parameter vary

Algorithm 3 Hybrid LX-SSA-VNS for a prescribed turbine count N

Require: budget B , split ρ , population N_p , LX-SSA parameters ϕ, χ , farm radius r

- 1: $T_1 \leftarrow \text{round}((\rho B - N_p)/(2N_p))$
- 2: Run LX-SSA (Algorithm 2) for T_1 iterations; let H be its food source
- 3: $x \leftarrow H$, $k \leftarrow 1$, step $\leftarrow 0.1r$
- 4: **while** fewer than B objective calls have been used **do**
- 5: $y \leftarrow x$ with k random turbines moved by $\mathcal{N}(0, (\sigma_k r)^2)$; box repair
- 6: **for** L local-search moves (while budget remains) **do**
- 7: $z \leftarrow y$ with one random turbine moved by $\mathcal{N}(0, \text{step}^2)$; box repair
- 8: **if** $F_p(z) < F_p(y)$ **then**
- 9: $y \leftarrow z$; step $\leftarrow \min(1.5 \text{ step}, 0.5r)$
- 10: **else**
- 11: step $\leftarrow \max(0.8 \text{ step}, 10^{-3}r)$
- 12: **end if**
- 13: **end for**
- 14: **if** $F_p(y) < F_p(x)$ **then**
- 15: $x \leftarrow y$; $k \leftarrow 1$
- 16: **else**
- 17: $k \leftarrow k \bmod k_{\max} + 1$
- 18: **end if**
- 19: **end while**
- 20: **return** x

with direction, from $\psi = 2.6$ (255 – 270°) to $\psi = 10$ (180 – 195°), with a broader directional distribution. Wind direction is discretized into $N_\theta = 24$ bins of 15° , and the speed range from $s_{\text{cut-in}}$ to s_{rated} into $N_s = 21$ intervals of 0.5 m/s.

Each data set is combined with three circular farms of radius $r = 500, 750$ and 1000 m. The turbine count is prescribed for

each run: $N = 2, \dots, 10$ for 500 m, $N = 2, \dots, 12$ for 750 m and $N = 2, \dots, 15$ for 1000 m. This gives $2 \times (9 + 11 + 14) = 68$ test cases. The turbine and wake parameters are

- rotor radius $R = 38.5$ m and minimum spacing $4D = 8R = 308$ m;
- cut-in speed $s_{\text{cut-in}} = 3.5$ m/s, rated speed $s_{\text{rated}} = 14$ m/s and rated power $P_{\text{rated}} = 1500$ kW;
- linear power-curve parameters $\lambda' = 140.86$ and $\eta = -500$;
- thrust coefficient $C_T = 0.8$ and wake spreading constant $K = 0.075$;
- constraint penalty $\lambda = 10^{10}$ in (7).

Every run optimizes the $2N$ turbine coordinates for its prescribed N ; varying N is a parameter sweep, not a joint optimization of turbine number and positions.

B. Methods Compared

The proposed hybrid and seven reference methods are compared, all with the same objective (7), the same bounds $[-r, r]^{2N}$ and the same budget:

- *LX-SSA-VNS* (proposed): Algorithm 3 with $\rho = 0.5$, i.e., 50 LX-SSA iterations (3,030 calls) followed by 3,000 calls of VNS.
- *LX-SSA* [30]: Algorithm 2 with $\phi = 0$, $\chi = 1$, population 30 and 100 iterations.
- *SSA* [26]: the same leader/follower dynamics without the Laplace candidate; population 30 and 200 iterations.
- *PSO*: inertia weight $w = 0.7$ and acceleration coefficients $c_1 = c_2 = 2$, with unbounded velocities and positions clipped to the bounds; population 30 and 200 iterations.
- *DE*: DE/rand/1/bin with $F = 0.5$ and $CR = 0.9$; population 30 and 200 iterations.
- *VNS* (modified; this study): a variant of basic VNS [22], [40] tailored to turbine coordinates. It starts from the best member of the initial population. Shaking moves $k \in \{1, 2, 3\}$ randomly chosen turbines by Gaussian steps with standard deviation $0.05r$, $0.15r$ and $0.40r$. Instead of a complete local search, each shake is followed by only ten first-improvement trial moves of single turbines with an adaptive step (enlarged by 1.5 after a success, reduced by 0.8 after a failure). An improvement is accepted and resets $k = 1$; otherwise k increases cyclically.
- *Multistart SLSQP* (*MS-SLSQP*), a gradient-based method: SciPy's sequential least-squares quadratic programming solver [43] minimizes the wake loss with the spacing and boundary constraints imposed explicitly and forward-difference gradients (1-m step). It starts from the members of the initial population in order of F_p and restarts until the budget is exhausted.
- *BVNS* (original basic VNS): the basic VNS of Mladenović and Hansen [40], [41] in the continuous form of Mladenović et al. [42], with the exterior penalty F_p for the constraints. It uses $k_{\text{max}} = 5$ nested ℓ_∞ neighbourhoods of the whole layout, $\rho_{k-1} < \|y - x\|_\infty \leq \rho_k$ with $\rho_k = 0.1kr$; shaking draws a point uniformly from the k -th neighbourhood, so all $2N$ coordinates move. It is followed by a complete best-improvement local search to

a local minimum: a derivative-free compass search evaluates all $4N$ moves $\pm h$ along the coordinates, moves to the best improving one, and halves h (from $0.05r$) when none improves, until $h < 10^{-3}r$. (Gradient descent, used for smooth problems in [42], is not applicable because the Jensen top-hat wake makes F_p discontinuous.) BVNS first applies this local search to the best member of the initial population; neighbourhood change is as in VNS, and the returned layout is the best point evaluated.

SSA, LX-SSA, PSO and DE use the original implementation. VNS, BVNS, MS-SLSQP and the hybrid were implemented for this study, and the hybrid calls the original LX-SSA code in its first phase. Comparing LX-SSA-VNS with LX-SSA and with stand-alone VNS isolates the contribution of each phase, and comparing VNS with BVNS shows the effect of the modifications made to the original VNS.

C. Budget, Seeds and Pairing

Every run has a budget of exactly 6,030 objective-function calls, counted by a wrapper around the objective. The count includes the initial population and, for MS-SLSQP, the calls used for finite-difference gradients. For LX-SSA this corresponds to 100 iterations, because each follower evaluates two candidates before the population is re-evaluated ($30 + 100 \times 60$ calls); SSA, PSO and DE reach the same count in 200 iterations ($30 + 200 \times 30$), and LX-SSA-VNS in $3,030 + 3,000$ calls. Each method is run with seeds 1–30. Every optimizer seeds the random-number generator and draws its initial population first, so runs with the same seed start from the *same* initial population for all eight methods. The runs of a test case are therefore paired by seed.

D. Performance Measures and Statistical Analysis

For each run we record

- the final benchmark objective and wake loss (as a percentage of the wake-free objective);
- whether the final layout is feasible (spacing and boundary satisfied within 10^{-6} m);
- the convergence curve, i.e., the best feasible objective found after every 30 calls;
- the wall-clock time.

Means and standard deviations are computed over the feasible runs.

Within each test case, the 30 seed-paired runs are compared by a run-level Friedman test and by two-sided Wilcoxon signed-rank tests of LX-SSA-VNS against each of the other seven methods. The seven p values of each case are adjusted by Holm's procedure [48], and effect sizes are reported as matched-pairs rank-biserial correlations. For ranking, a feasible run beats an infeasible one, and infeasible runs are compared by their spacing shortfall. Across the 68 cases, the methods are ranked within each case by their mean feasible objective and compared by a case-level Friedman test with the Iman–Davenport correction and Holm-adjusted average-rank post hoc tests, following Demšar [47].

TABLE II: Case-level Friedman analysis over all 68 benchmark cases (blocks: cases; ranks of the mean feasible objective, 1 = best). Friedman $\chi_F^2 = 212.5$ (7 d.f.), $p = 2.57 \times 10^{-42}$; Iman–Davenport $F_F = 54.0$. Holm-adjusted p values of the average-rank z test against the best-ranked method (VNS) and against LX-SSA-VNS; “Best”: cases in which the method alone has the highest mean (remaining cases are ties); “Feas.”: percentage of feasible runs.

Method	Avg. rank	Best	Feas. (%)	p vs. VNS	p vs. LX-SSA-VNS
VNS	2.09	36	99.6	–	0.123
LX-SSA-VNS	2.74	5	98.1	0.123	–
BVNS	3.62	10	100.0	5.08×10^{-4}	0.068
SSA	4.79	0	97.6	3.55×10^{-10}	2.86×10^{-6}
PSO	4.84	3	86.0	2.36×10^{-10}	2.22×10^{-6}
MS-SLSQP	5.38	6	99.9	2.22×10^{-14}	1.48×10^{-9}
LX-SSA	5.68	0	96.9	7.94×10^{-17}	1.52×10^{-11}
DE	6.86	0	71.4	4.64×10^{-29}	6.50×10^{-22}

E. Implementation Verification

The objective function was vectorized and checked against the original implementation; the relative difference is below 10^{-6} on random feasible and infeasible layouts. Its wake and energy terms also reproduce the objective values of 720 archived final layouts from an earlier run of the original code (maximum absolute deviation 8.7×10^{-11}). Re-running the original SSA, LX-SSA, PSO and DE code at the archived settings reproduces the archived distributions: 22 of 24 Mann–Whitney comparisons give $p \geq 0.05$, and the means differ by at most 0.21%. This also confirms the Laplace parameters $\phi = 0$ and $\chi = 1$.

F. Computing Environment

All runs were executed on a Linux container with an Intel Xeon processor at 2.80 GHz and 4 virtual cores, four runs in parallel, using Python 3.11.15, NumPy 2.4.6 and SciPy 1.17.1. Each run was timed individually. Scripts, per-run results (including final coordinates and convergence curves) and the figure-generation code accompany the revision.

VIII. NUMERICAL RESULTS

This section reports 16,320 runs: 68 test cases, eight methods and 30 seed-paired runs, all at 6,030 objective calls. It is followed by a comparison of the original and modified VNS, an ablation of the hybrid and the Horns Rev 1 site case. Detailed per-case results are given in Tables IV–IX.

A. Overall Comparison

The case-level Friedman test over the 68 cases rejects equal performance of the eight methods ($\chi_F^2 = 212.5$, 7 d.f., $p = 2.6 \times 10^{-42}$; Iman–Davenport $F_F = 54.0$). Table II and Fig. 4 give the average ranks (in parentheses: cases in which the method alone has the highest mean):

- VNS (modified) 2.09 (36 cases);
- **LX-SSA-VNS 2.74** (5 cases);
- BVNS (original VNS) 3.62 (10 cases);
- SSA 4.79, PSO 4.84, MS-SLSQP 5.38, LX-SSA 5.68 and DE 6.86.

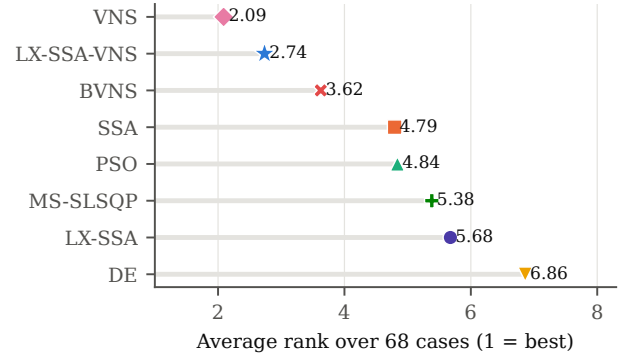


Fig. 4: Average rank of the eight methods over the 68 benchmark cases (ranks of the mean feasible objective within each case; 1 = best).

TABLE III: Pairwise outcome of the hybrid LX-SSA-VNS against each method: number of cases in which LX-SSA-VNS is significantly better / not significantly different / significantly worse (two-sided Wilcoxon signed-rank test on 30 seed-paired runs, Holm-adjusted over the seven comparisons of each case, $\alpha = 0.05$).

	DS	r (m)	Cases	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP	BVNS
I	500	9	3/6/0	0/9/0	3/6/0	6/2/1	0/7/2	3/4/2	2/4/3	
I	750	11	8/3/0	5/6/0	4/7/0	9/2/0	0/10/1	7/4/0	4/7/0	
I	1000	14	6/8/0	7/7/0	8/6/0	12/2/0	0/9/5	10/4/0	2/12/0	
II	500	9	4/5/0	2/7/0	4/5/0	6/3/0	0/7/2	2/4/3	1/6/2	
II	750	11	6/5/0	5/6/0	5/6/0	8/3/0	0/9/2	3/8/0	1/9/1	
II	1000	14	8/6/0	8/6/0	8/6/0	11/3/0	0/11/3	9/5/0	2/11/1	
All		68	35/33/0	27/41/0	32/36/0	52/15/1	0/53/15	34/29/5	12/49/7	

In the Holm-adjusted post hoc tests, LX-SSA-VNS ranks significantly better than SSA, PSO, MS-SLSQP, LX-SSA and DE ($p_{\text{Holm}} \leq 2.9 \times 10^{-6}$). Its differences from VNS ($p_{\text{Holm}} = 0.123$) and from BVNS ($p_{\text{Holm}} = 0.068$) are not significant at the case level, whereas VNS ranks significantly better than BVNS ($p_{\text{Holm}} = 5.1 \times 10^{-4}$).

The run-level pairwise tests (Table III) show where the hybrid gains and loses:

- **LX-SSA-VNS vs. LX-SSA.** The hybrid is significantly better in 35 cases and never worse (r_{tb} from 0.50 to 0.98, median 0.77). The 33 cases without a significant difference are mostly small cases ($N \leq 7$), in which LX-SSA is already close to the best layouts, plus four cases with $N = 8$ or 10, including both 500-m ten-turbine cases, in which many runs of both methods are infeasible.
- **LX-SSA-VNS vs. SSA and PSO.** The hybrid is significantly better in 27 and 32 cases, respectively, and never worse.
- **LX-SSA-VNS vs. DE.** The hybrid is better in 52 cases and worse in only one three-turbine case.
- **LX-SSA-VNS vs. MS-SLSQP.** The hybrid is better in 34 cases and worse in five, all dense 500-m cases (Data Set I, $N = 9, 10$; Data Set II, $N = 8-10$), where MS-SLSQP’s explicit constraint handling pays off.
- **LX-SSA-VNS vs. VNS.** The two are statistically indistinguishable in 53 cases. VNS is significantly better in 15

cases, all with $N \geq 7$ (median $r_{rb} = -0.62$), and the hybrid is never significantly better than VNS.

- *LX-SSA-VNS vs. BVNS*. The hybrid is significantly better in 12 cases, all with $N = 5$ to 11, and worse in seven: five cases with $N = 3$ or 4, and the two 500-m ten-turbine cases, in which BVNS is feasible in every run.

Hybridization therefore lifts LX-SSA from the seventh to the second average rank among the eight methods, but it does not surpass the modified VNS at this budget.

B. Solution Quality by Farm Size and Turbine Count

Figure 5 and Tables IV–IX show that the wake loss grows with N , falls with the farm radius, and is higher for the broader wind rose of Data Set II. For $N \leq 3$ all methods stay within 0.3% of the wake-free objective. With growing N the curve of the hybrid separates from that of LX-SSA and moves close to the VNS and MS-SLSQP curves, as in these examples:

- Data Set I, 1000 m/15: LX-SSA-VNS 2.08%, VNS 1.82%, BVNS 1.99%, MS-SLSQP 2.20%, SSA 2.80%, LX-SSA 3.32%;
- Data Set II, 750 m/12: LX-SSA-VNS 5.95%, VNS 5.76%, BVNS 6.17%, MS-SLSQP 5.60%, SSA 6.79%, LX-SSA 7.34%.

In every largest case the hybrid reduces the mean wake loss of LX-SSA by 1.2 to 2.0 percentage points, while VNS or MS-SLSQP retains the lowest value.

C. Feasibility

Figure 6 shows that LX-SSA-VNS returns a feasible layout in 98.1% of all runs, compared with 96.9% for LX-SSA, 97.6% for SSA, 86.0% for PSO, 71.4% for DE, 99.6% for VNS and 99.9% for MS-SLSQP. BVNS is the only method that is feasible in every run. The only cases in which the hybrid misses feasibility are the densest 500-m cases: 26 of 30 runs are feasible for $N = 9$ and 15 of 30 for $N = 10$, compared with 21 and 13 for LX-SSA, 29 and 27 for VNS, and 30 and 30 for BVNS. In these cases the LX-SSA phase frequently ends with an infeasible layout, and the VNS phase, which works on a single incumbent, cannot always repair it in the remaining budget. The feasibility counts of all penalty-based methods are identical for the two wind data sets, because while every candidate is infeasible the penalty term dominates F_p .

D. Convergence Behaviour

Figures 7 and 8 show the effect of the hybridization directly. Until 3,030 calls the hybrid runs a shortened LX-SSA and is therefore slightly behind the full LX-SSA run. After the switch its curve drops steeply, as VNS continues to improve the incumbent that LX-SSA has found, whereas stand-alone LX-SSA and SSA stagnate after about 3,000 calls. Over all feasible runs, the VNS phase removes on average 36.4% (median 31.1%) of the wake loss present at the end of the LX-SSA phase, and it turns 82 runs that were still infeasible at the switch into feasible ones. Stand-alone VNS reaches low wake losses earlier because it spends its whole budget on intensification, and MS-SLSQP is the fastest method to

reach feasibility. BVNS reaches feasibility in the dense 500-m cases earlier than VNS, but afterwards it improves in fewer and larger steps, because every sweep of its complete local search costs $4N$ calls.

E. Distribution of Results

Figure 9 shows the run-to-run distributions for the moderate cases. LX-SSA-VNS has the lowest median wake loss in both 500-m cases and the second-lowest, after VNS, in the other four, with a smaller interquartile range than LX-SSA in five of the six cases. DE has the highest median in all six cases.

F. Original versus Modified VNS

The VNS used as a baseline and inside the hybrid departs from the original algorithm in two ways: it perturbs only one to three turbines, and it replaces the complete local search by ten trial moves. Table X shows how the original basic VNS (BVNS), with shaking of the whole layout and a complete best-improvement local search, compares with every other method.

- *BVNS vs. the population-based methods and MS-SLSQP*. BVNS is significantly better than LX-SSA in 32 cases, SSA in 24, PSO in 30, DE in 53 and MS-SLSQP in 24, and worse in at most four cases against any of them.
- *BVNS vs. the modified VNS*. BVNS is better in eight cases, all with $N = 3$ or 4, and worse in 27 cases, all with $N \geq 6$ (median $r_{rb} = -0.78$). For small layouts a complete descent over all coordinate moves is affordable and pays off; for larger layouts every sweep costs $4N$ calls, so few VNS iterations fit into the budget, and the cheaper turbine-wise moves of the modified VNS make better use of it.
- *BVNS vs. LX-SSA-VNS*. BVNS is better in eight cases and worse in 11. These counts differ slightly from Table III because the Holm adjustment is applied to a different family of comparisons.
- *Feasibility*. BVNS is the only method that returns a feasible layout in all 2,040 runs, including the dense 500-m cases.

The modifications therefore make VNS stronger on the larger layouts that matter most in practice, which supports using the modified VNS as the intensification phase of the hybrid; the original BVNS nevertheless remains a strong baseline.

G. Ablation of the Hybrid

Two ablations clarify the role of each phase:

- *Phase contribution*. Comparing LX-SSA-VNS with LX-SSA (same first phase) and with VNS (same second phase) shows that nearly all of the improvement over LX-SSA comes from the VNS phase: the hybrid beats LX-SSA in 35 cases and never loses. The LX-SSA phase, however, does not provide a better starting point than the best member of a random initial population, which stand-alone VNS uses: stand-alone VNS remains better in 15 dense cases.

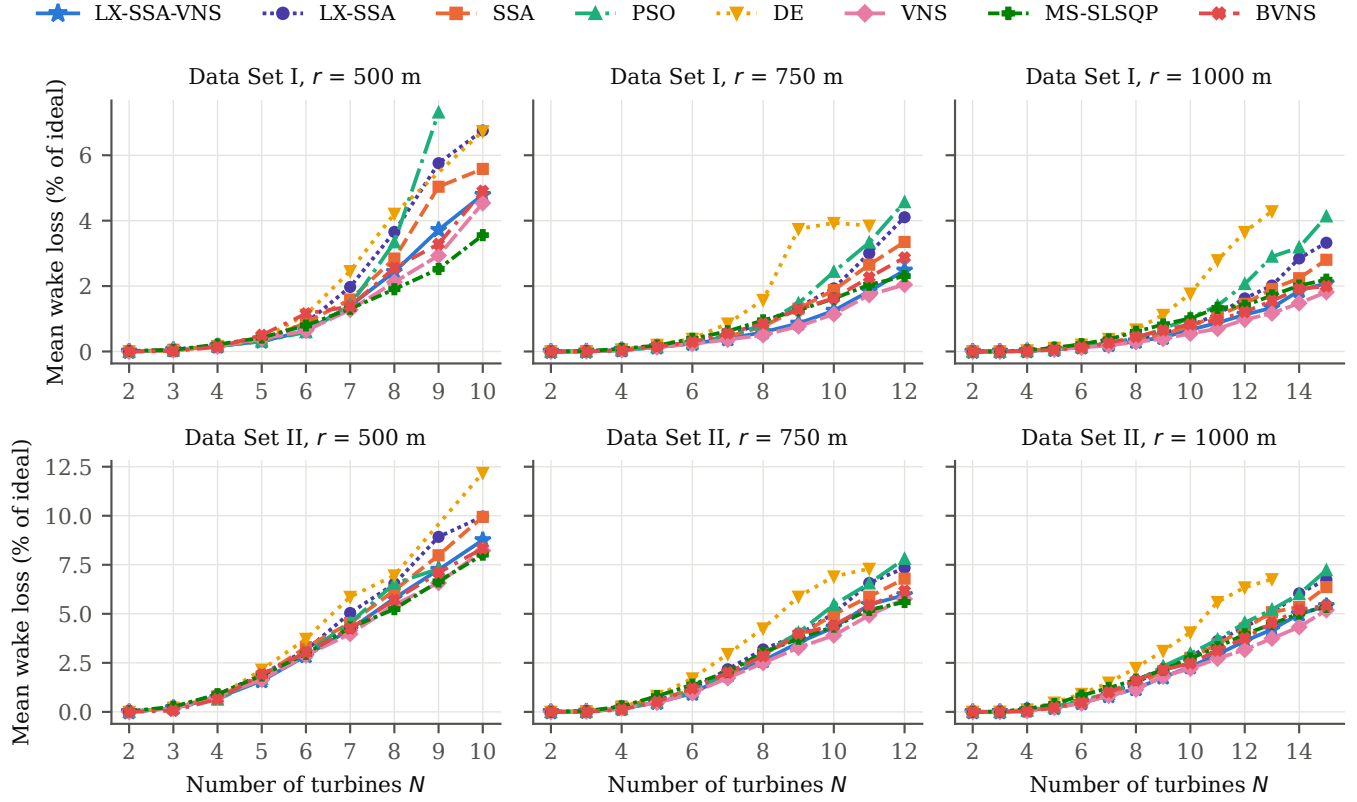


Fig. 5: Mean wake loss (percentage of the wake-free objective) of the feasible runs versus the prescribed number of turbines; a point is omitted when a method has no feasible run.

TABLE IV: Data Set I, 500-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	LX-SSA-VNS	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP	BVNS
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28087.2 (4.6)	28091.5 (0.0)
3	42137.2	42117.1 (13.6)	42114.8 (14.2)	42114.9 (10.3)	42124.7 (8.5)	42130.7 (8.8)	42119.7 (11.8)	42116.0 (10.1)	42131.6 (9.7)
4	56182.9	56092.3 (22.6)	56078.8 (22.6)	56086.8 (19.8)	56094.3 (13.3)	56092.8 (29.8)	56094.2 (28.5)	56063.2 (30.9)	56106.1 (21.7)
5	70228.7	70008.9 (40.7)	69972.6 (72.0)	69994.0 (34.0)	70010.5 (27.2)	69960.5 (78.6)	69955.1 (126.5)	69930.5 (113.0)	69876.9 (199.8)
6	84274.4	83748.8 (232.3)	83544.9 (465.3)	83535.6 (372.5)	83775.8 (154.1)	83326.7 (761.5)	83738.2 (213.4)	83601.4 (213.1)	83299.3 (524.5)
7	98320.2	96944.1 (569.8)	96380.6 (1013.8)	96757.0 (692.4)	96935.0 (485.2)	95903.3 (1316.3) ¹⁵	97066.5 (484.7)	97037.0 (484.3)	96967.1 (590.6)
8	112365.9	109629.4 (1262.9)	108258.3 (1704.3) ²⁸	109172.6 (1362.4)	108616.6 (1480.8) ²⁴	107632.7 (1592.8) ²	109981.7 (835.2)	110226.6 (585.5)	109493.5 (2035.2)
9	126411.6	121711.4 (1326.2) ²⁶	119134.2 (2666.4) ²¹	120043.1 (2755.8) ²⁵	117175.2 (7148.7) ²	–	122702.8 (1513.8) ²⁹	123222.8 (812.6)²⁸	122259.2 (1232.0)
10	140457.4	133731.1 (1839.9) ¹⁵	130967.4 (3191.0) ¹³	132616.3 (2328.0) ¹²	–	130994.8 (0.0) ¹	134084.5 (1603.0) ²⁷	135461.0 (1291.9)	133567.5 (1820.9)

TABLE V: Data Set I, 750-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	LX-SSA-VNS	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP	BVNS
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.0 (1.6)	28091.5 (0.0)
3	42137.2	42135.7 (3.5)	42134.2 (4.8)	42134.4 (4.1)	42137.0 (1.1)	42136.8 (1.5)	42134.5 (4.9)	42132.3 (5.0)	42137.2 (0.0)
4	56182.9	56157.8 (15.0)	56149.9 (14.0)	56153.8 (17.9)	56160.0 (14.2)	56146.9 (17.5)	56156.3 (14.4)	56137.1 (16.4)	56167.0 (13.8)
5	70228.7	70135.1 (36.5)	70118.1 (28.6)	70122.2 (26.3)	70135.0 (22.6)	70083.9 (31.7)	70137.4 (36.3)	70100.5 (42.3)	70125.8 (47.0)
6	84274.4	84070.5 (38.2)	84033.5 (58.3)	84035.2 (58.4)	84059.1 (42.0)	83948.2 (74.1)	84079.8 (64.2)	83947.6 (95.9)	84029.2 (141.3)
7	98320.2	97939.2 (105.7)	97840.0 (167.7)	97875.2 (169.3)	97872.3 (131.8)	97479.1 (285.1)	97963.6 (80.6)	97710.2 (204.8)	97774.4 (232.6)
8	112365.9	111702.9 (228.8)	111374.0 (460.8)	111506.5 (334.5)	111448.4 (551.6)	110595.8 (445.6)	111801.5 (162.3)	111296.2 (217.8)	111411.4 (493.5)
9	126411.6	125328.2 (407.8)	124698.0 (960.4)	124756.2 (610.5)	124550.0 (794.2)	121674.9 (2276.5) ²³	125437.8 (379.2)	124818.4 (466.3)	124780.0 (716.8)
10	140457.4	138702.3 (605.3)	137748.6 (1116.3)	137817.6 (1160.1)	137036.8 (1051.3) ²⁸	134937.7 (998.5) ⁴	138856.6 (431.7)	138184.7 (595.3)	138170.4 (764.7)
11	154503.1	151656.0 (753.7)	149863.8 (1886.6) ²⁹	150404.1 (1042.2) ²⁹	149356.6 (1849.0) ²⁰	148555.7 (1134.8) ³	151831.7 (908.0)	151352.9 (560.2)	151004.2 (1339.0)
12	168548.8	164372.6 (1167.7)	161630.5 (1988.2) ²⁸	162903.5 (1685.8)	160854.7 (2108.4) ⁷	–	165106.7 (762.7)	164663.8 (674.6)	163712.4 (1575.6)

• *Budget split.* Table XI varies ρ over 25%, 50% and 75%

on twelve representative cases. Giving more budget to

TABLE VI: Data Set I, 1000-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

<i>N</i>	Ideal	LX-SSA-VNS	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP	BVNS
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.3 (0.7)	28091.5 (0.0)
3	42137.2	42136.6 (2.4)	42136.7 (1.4)	42136.7 (1.6)	42137.2 (0.0)	42137.2 (0.0)	42137.0 (0.9)	42135.9 (2.6)	42137.2 (0.0)
4	56182.9	56173.6 (7.5)	56172.9 (7.8)	56169.4 (9.2)	56173.0 (6.8)	56166.9 (8.4)	56170.1 (13.3)	56162.5 (10.5)	56177.3 (8.6)
5	70228.7	70189.3 (14.8)	70183.3 (21.0)	70174.0 (19.7)	70182.6 (13.0)	70148.7 (22.4)	70186.7 (23.4)	70151.4 (32.7)	70193.3 (24.5)
6	84274.4	84160.5 (32.4)	84142.7 (23.5)	84149.9 (29.7)	84152.3 (22.2)	84096.1 (24.0)	84171.1 (30.2)	84087.4 (92.6)	84170.0 (49.5)
7	98320.2	98114.3 (54.2)	98078.4 (82.1)	98084.7 (47.0)	98083.0 (35.8)	97942.3 (67.9)	98141.0 (51.2)	97955.8 (136.6)	98064.6 (118.7)
8	112365.9	112018.1 (104.7)	111924.6 (164.7)	111947.1 (125.7)	111915.5 (125.3)	111608.3 (168.9)	112038.8 (78.6)	111688.0 (181.9)	111862.5 (246.1)
9	126411.6	125870.2 (134.6)	125701.5 (265.4)	125761.1 (168.4)	125531.1 (324.5)	124996.3 (282.9)	125923.2 (134.4)	125366.6 (282.7)	125589.7 (312.6)
10	140457.4	139526.2 (256.3)	139354.2 (388.5)	139348.0 (433.0)	139018.6 (478.6)	137969.0 (623.0)	139687.2 (221.5)	139031.6 (341.6)	139290.3 (447.7)
11	154503.1	153147.5 (320.5)	152736.8 (558.8)	152825.1 (385.8)	152337.8 (768.8) ²⁹	150189.7 (1068.9) ²⁹	153426.4 (348.5)	152445.9 (402.0)	153010.5 (393.8)
12	168548.8	166651.8 (513.8)	165814.4 (827.9)	166091.9 (816.6)	165072.5 (1367.5) ²⁸	162390.4 (1765.9) ¹⁷	166906.5 (459.2)	166188.4 (681.0)	166510.6 (669.8)
13	182594.6	180156.4 (701.3)	178913.6 (967.4)	179101.5 (1028.9)	177302.0 (2352.6) ²²	174763.4 (1389.6) ⁴	180466.7 (599.8)	179465.2 (506.2)	179762.6 (1091.2)
14	196640.3	192979.6 (804.0)	191046.6 (1754.6)	192235.7 (1504.8)	190385.1 (1544.4) ¹⁸	– ⁰	193732.7 (887.0)	192677.8 (595.4)	192874.6 (1745.6)
15	210686.1	206293.9 (874.2)	203681.8 (1899.5) ²⁹	204777.6 (1420.5)	201985.5 (1678.3) ⁹	– ⁰	206859.5 (910.3)	206059.6 (761.8)	206486.7 (1093.6)

TABLE VII: Data Set II, 500-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

<i>N</i>	Ideal	LX-SSA-VNS	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP	BVNS
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14626.0 (5.6)	14630.8 (0.0)
3	21946.1	21901.3 (11.1)	21899.7 (9.5)	21900.9 (11.1)	21901.5 (6.3)	21901.2 (5.5)	21906.5 (23.0)	21888.8 (45.5)	21932.3 (17.2)
4	29261.5	29052.1 (72.4)	29004.2 (87.4)	29028.7 (93.8)	29077.9 (64.5)	29044.2 (61.5)	29028.5 (66.7)	28998.9 (104.2)	29070.8 (94.1)
5	36576.9	35996.8 (137.2)	35880.1 (168.6)	35918.9 (74.0)	35959.4 (119.0)	35789.6 (193.8)	35976.5 (173.2)	35896.9 (110.0)	35878.2 (160.9)
6	43892.3	42635.1 (202.0)	42512.1 (227.9)	42444.3 (224.5)	42568.5 (159.7)	42257.5 (374.9)	42626.6 (192.4)	42590.7 (236.0)	42560.3 (202.5)
7	51207.6	49055.4 (161.0)	48626.5 (340.5)	48887.9 (304.1)	48885.9 (278.3)	48203.9 (767.8) ¹⁵	49171.8 (236.9)	49007.9 (208.2)	49032.1 (314.4)
8	58523.0	55132.6 (337.9)	54699.5 (620.7) ²⁸	54900.5 (471.3)	54693.1 (404.3) ²⁴	54451.3 (618.7) ²	55331.0 (323.5)	55457.2 (287.3)	55182.8 (738.6)
9	65838.4	61066.7 (640.2) ²⁶	59963.3 (652.2) ²¹	60581.0 (844.8) ²⁵	61035.6 (137.9) ²	– ⁰	61509.3 (538.6)²⁹	61457.0 (370.7)	61166.8 (574.8)
10	73153.8	66744.8 (731.7) ¹⁵	65868.7 (586.0) ¹³	65880.7 (761.6) ¹²	– ⁰	64240.2 (0.0) ¹	67141.9 (581.4) ²⁷	67288.3 (300.1)²⁹	67056.0 (749.1)

TABLE VIII: Data Set II, 750-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

<i>N</i>	Ideal	LX-SSA-VNS	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP	BVNS
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.4 (1.5)	14630.8 (0.0)
3	21946.1	21941.4 (6.9)	21937.3 (8.1)	21940.0 (8.2)	21943.2 (5.4)	21942.5 (5.8)	21941.5 (6.8)	21938.2 (12.2)	21946.1 (0.0)
4	29261.5	29214.1 (25.3)	29201.4 (25.7)	29182.5 (36.2)	29213.0 (20.4)	29195.3 (26.2)	29202.4 (32.5)	29179.0 (59.2)	29227.3 (29.9)
5	36576.9	36402.3 (84.4)	36366.8 (74.3)	36332.0 (91.4)	36400.0 (62.2)	36277.7 (105.9)	36405.5 (82.8)	36283.6 (98.0)	36389.7 (97.1)
6	43892.3	43477.6 (122.5)	43377.4 (175.7)	43419.5 (153.1)	43441.8 (107.5)	43142.2 (144.8)	43470.4 (120.0)	43283.1 (153.3)	43367.4 (201.2)
7	51207.6	50250.7 (154.8)	50095.4 (193.6)	50264.8 (202.5)	50188.2 (199.5)	49696.6 (156.3)	50324.3 (194.4)	50164.7 (205.9)	50180.2 (303.8)
8	58523.0	56999.0 (223.5)	56659.2 (332.0)	56806.2 (303.6)	56786.4 (311.8)	56033.1 (331.4)	57063.0 (286.7)	56752.4 (256.7)	56855.9 (391.5)
9	65838.4	63513.2 (260.8)	63252.7 (342.8)	63291.6 (402.5)	63180.2 (411.7)	61970.8 (763.0) ²³	63680.4 (415.5)	63381.6 (322.9)	63217.0 (515.4)
10	73153.8	70004.6 (370.2)	69480.7 (385.7)	69520.3 (512.6)	69159.6 (436.1) ²⁸	68103.2 (423.7) ⁴	70300.6 (409.0)	69972.9 (458.1)	69930.2 (355.0)
11	80469.2	76071.3 (533.6)	75181.6 (681.1) ²⁹	75742.5 (677.1) ²⁹	75199.6 (585.7) ²⁰	74596.6 (476.3) ³	76504.5 (395.7)	76284.4 (433.9)	76099.7 (566.3)
12	87784.5	82557.7 (550.8)	81341.8 (792.1) ²⁸	81822.1 (625.7)	80936.3 (798.4) ⁷	– ⁰	82731.2 (470.9)	82867.2 (399.2)	82368.6 (603.1)

LX-SSA is harmful: the 50% split has a higher mean than the 75% split in 11 of 12 cases, significantly so in four. Giving more budget to VNS helps slightly: the 25% split has the higher mean in 9 of 12 cases, but the difference is significant in only one.

Both ablations point the same way: at this budget the intensification phase is the valuable part of the hybrid. A larger total budget, or a switch triggered by stagnation of the swarm instead of a fixed split, would be needed for the exploration phase to pay off.

H. Optimized Layouts

Figure 10 compares, for the largest turbine count of each farm, the best LX-SSA-VNS layout with the best layout found

by any method, which comes from BVNS in three cases, from MS-SLSQP in two and from VNS in one. Most turbines are placed close to the boundary, which maximizes their mutual distances, while the remaining turbines occupy the interior at positions staggered with respect to the dominant wind directions.

I. Computational Cost

At an equal number of objective calls all methods need almost the same time: 0.37 ms per call for BVNS, 0.39 ms for LX-SSA-VNS, 0.40 ms for LX-SSA and SSA, 0.41 ms for VNS and PSO, 0.47 ms for DE and 0.49 ms for MS-SLSQP. The objective evaluation dominates the cost, so neither the Laplace step nor the VNS bookkeeping adds measurable

TABLE IX: Data Set II, 1000-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	LX-SSA-VNS	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP	BVNS
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.5 (1.0)	14630.8 (0.0)
3	21946.1	21945.8 (1.2)	21945.3 (2.8)	21945.6 (1.8)	21945.8 (2.0)	21946.1 (0.0)	21945.6 (1.7)	21943.5 (4.8)	21946.1 (0.0)
4	29261.5	29236.8 (17.4)	29228.0 (14.9)	29232.5 (18.5)	29243.3 (13.6)	29225.9 (12.1)	29246.6 (13.5)	29218.7 (24.2)	29253.1 (11.7)
5	36576.9	36499.5 (32.4)	36468.2 (46.1)	36485.9 (37.5)	36486.0 (36.0)	36402.8 (39.4)	36499.6 (38.8)	36432.0 (50.4)	36498.4 (49.1)
6	43892.3	43688.5 (69.2)	43656.6 (73.3)	43626.4 (74.5)	43665.8 (65.6)	43488.5 (69.7)	43700.2 (82.3)	43524.8 (86.8)	43686.1 (101.2)
7	51207.6	50786.6 (120.8)	50727.3 (119.3)	50731.8 (118.8)	50739.2 (134.6)	50442.7 (136.0)	50785.0 (110.8)	50582.5 (165.5)	50696.0 (203.8)
8	58523.0	57832.5 (140.8)	57645.1 (186.6)	57739.3 (142.0)	57616.2 (157.7)	57210.0 (191.8)	57854.1 (163.9)	57560.8 (206.6)	57588.6 (311.3)
9	65838.4	64672.1 (262.3)	64428.7 (273.1)	64476.4 (203.0)	64308.4 (234.8)	63790.1 (227.2)	64652.4 (279.3)	64419.1 (268.8)	64430.8 (311.5)
10	73153.8	71475.5 (282.8)	71099.4 (266.9)	71139.4 (336.8)	70985.2 (314.4)	70190.1 (286.9)	71535.7 (321.0)	71161.4 (275.8)	71342.3 (339.4)
11	80469.2	78115.5 (396.3)	77561.6 (498.0)	77697.5 (425.8)	77501.2 (352.1) ²⁹	75960.9 (577.6) ²⁹	78281.9 (285.9)	77812.4 (323.7)	77948.1 (483.3)
12	87784.5	84615.4 (456.0)	83981.2 (515.0)	84049.6 (660.3)	83795.7 (565.1) ²⁸	82211.6 (905.2) ¹⁷	85002.5 (322.8)	84312.7 (466.3)	84485.7 (606.8)
13	95099.9	91106.0 (521.1)	90303.5 (674.2)	90267.0 (482.6)	90139.1 (533.9) ²²	88665.1 (496.0) ⁴	91538.1 (357.6)	90837.9 (572.0)	90787.8 (806.9)
14	102415.3	97312.5 (738.6)	96229.6 (1019.3)	96936.5 (669.3)	96266.2 (730.7) ¹⁸	– ⁰	97983.6 (409.3)	97276.4 (405.3)	97079.1 (980.3)
15	109730.7	103802.7 (659.0)	102327.4 (838.7) ²⁹	102746.3 (827.8)	101804.8 (571.6) ⁹	– ⁰	104036.5 (619.7)	103853.3 (475.2)	103777.6 (794.4)

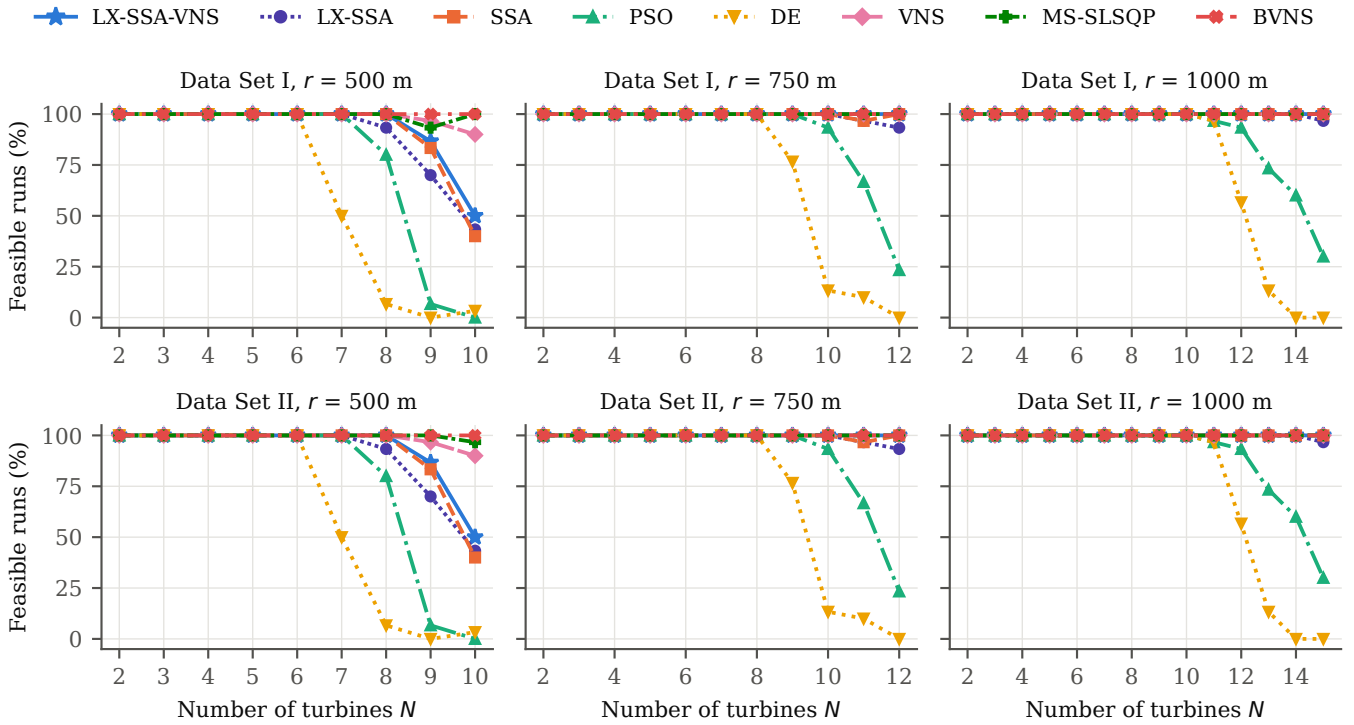


Fig. 6: Percentage of the 30 runs that end with a feasible layout (all turbines inside the farm and at least 308 m apart).

overhead. A 6,030-call run of LX-SSA-VNS takes 1.7–3.5 s on average, depending on N .

J. Horns Rev 1 Site Case

To bring the validation closer to a real project, the Horns Rev 1 offshore farm was modeled with its real turbine, wind climate and site outline, all taken from the PyWake distribution [44]:

- the installed layout of 80 Vestas V80 turbines (rotor diameter 80 m, 2 MW) and the V80 power and thrust-coefficient curves;
- the measured 12-sector wind climate;
- the parallelogram spanned by the installed turbines as the site outline.

The farm model keeps the Jensen top-hat wake (with $k = 0.04$, a value commonly used offshore) and root-sum-square superposition. It uses the tabulated power curve with a 25 m/s cut-out, a speed-dependent thrust coefficient at the free-stream speed, 5° direction bins and 1 m/s speed bins. For the installed layout it gives 664.6 GWh/yr and a wake loss of 10.95%, against 662.5 GWh/yr and 10.96% from PyWake’s Jensen model with the same settings, a difference below 0.3%. The 16-turbine block that PyWake uses as an example (30 seeds) and the complete 80-turbine farm (10 seeds) were optimized with all eight methods at 6,030 calls and the 4D (320 m) minimum spacing.

Table XII and Figs. 11 and 12 lead to three observations.

- *The benchmark ordering carries over to the site.*

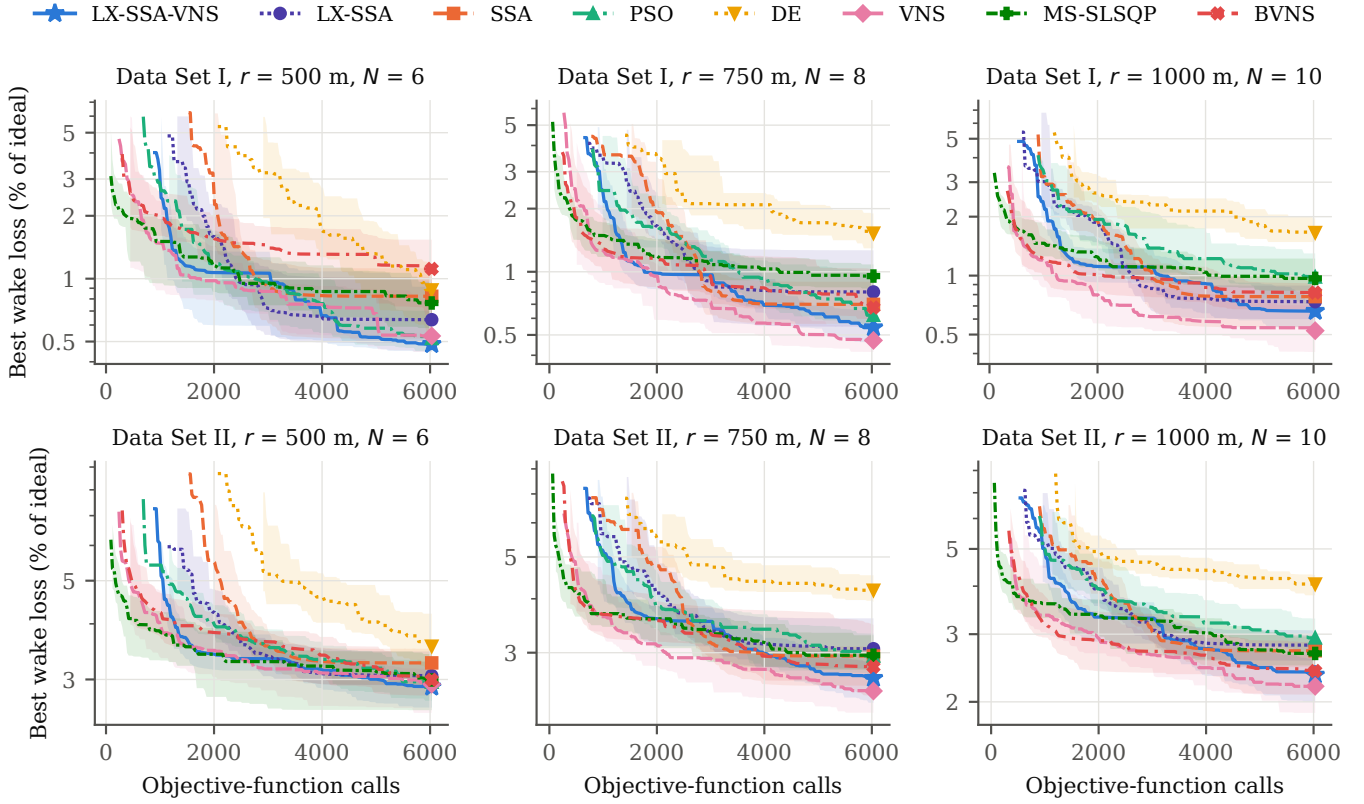


Fig. 7: Convergence for moderate turbine counts: median (line) and interquartile range (band) of the best feasible wake loss found so far over the 30 runs, versus the number of objective calls (logarithmic scale). A curve starts once at least half of the runs have found a feasible layout; the LX-SSA-VNS curve switches from the LX-SSA to the VNS phase at 3,030 calls.

TABLE X: Pairwise outcome of the original basic VNS (BVNS) against each method: number of cases in which BVNS is significantly better / not significantly different / significantly worse (two-sided Wilcoxon signed-rank test on 30 seed-paired runs, Holm-adjusted over the seven comparisons of each case, $\alpha = 0.05$).

DS	r (m)	Cases	LX-SSA-VNS	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP
I	500	9	3/4/2	5/4/0	4/5/0	4/3/2	4/5/0	1/7/1	3/4/2
I	750	11	0/7/4	4/7/0	3/8/0	3/8/0	9/2/0	2/3/6	3/7/1
I	1000	14	1/11/2	6/8/0	4/9/1	7/7/0	12/2/0	1/5/8	8/6/0
II	500	9	2/6/1	6/3/0	3/6/0	4/5/0	6/3/0	1/8/0	3/6/0
II	750	11	1/10/0	5/6/0	6/5/0	5/6/0	10/1/0	2/4/5	3/7/1
II	1000	14	1/11/2	6/8/0	4/10/0	7/7/0	12/2/0	1/6/7	4/10/0
All		68	8/49/11	32/36/0	24/43/1	30/36/2	53/15/0	8/33/27	24/40/4

TABLE XI: Sensitivity of LX-SSA-VNS to the budget split between the LX-SSA and VNS phases (25%, 50% and 75% of the 6,030 calls for LX-SSA): mean benchmark objective of the feasible runs (superscript: feasible runs when fewer than 30) and Holm-adjusted Wilcoxon signed-rank p of the 50% split against the 25% and 75% splits.

DS	r	N	25%	50%	75%	p (25)	p (75)
1	500	6	83714.3	83748.8	83696.8	0.904	0.904
1	500	10	134508.4 ¹⁷	133731.1 ¹⁵	133245.1 ¹¹	0.556	0.411
1	750	8	111722.3	111702.9	111666.2	0.832	0.832
1	750	12	164712.0 ²⁹	164372.6	163486.4 ²⁹	0.205	0.121
1	1000	10	139670.7	139526.2	139402.5	0.012	0.221
1	1000	15	206690.1	206293.9	205669.4 ²⁹	0.077	0.077
2	500	6	42581.2	42635.1	42481.3	0.477	0.017
2	500	10	66934.0 ¹⁷	66744.8 ¹⁵	66541.7 ¹⁰	0.670	0.168
2	750	8	57076.0	56999.0	57010.2	0.354	0.871
2	750	12	82679.2 ²⁹	82557.7	82102.7 ²⁹	0.855	0.013
2	1000	10	71434.4	71475.5	71193.5	0.280	0.009
2	1000	15	104084.1	103802.7	103307.0 ²⁹	0.077	0.006

In the 16-turbine block VNS attains the highest mean AEP (137.97 GWh/yr), followed by BVNS (137.67 GWh/yr, feasible in all 30 runs) and LX-SSA-VNS (137.59 GWh/yr). The hybrid is significantly better than LX-SSA ($p_{\text{Holm}} = 0.003$) and MS-SLSQP ($p_{\text{Holm}} = 0.040$); its differences from SSA and VNS ($p_{\text{Holm}} = 0.059$ each) and from BVNS ($p_{\text{Holm}} = 0.191$) are not significant. PSO and DE never find a feasible layout inside the parallelogram.

- No method improves on the installed layout within the

budget. The installed 7D grid reaches 139.51 GWh/yr.

- The full farm exceeds what the penalty-based methods can handle at this budget. With 160 variables and 3,160 spacing constraints, none of the runs of LX-SSA-VNS, LX-SSA, SSA, PSO, DE, VNS or BVNS returns a feasible layout. MS-SLSQP, which handles the constraints explicitly, is feasible in all ten runs, but its mean AEP

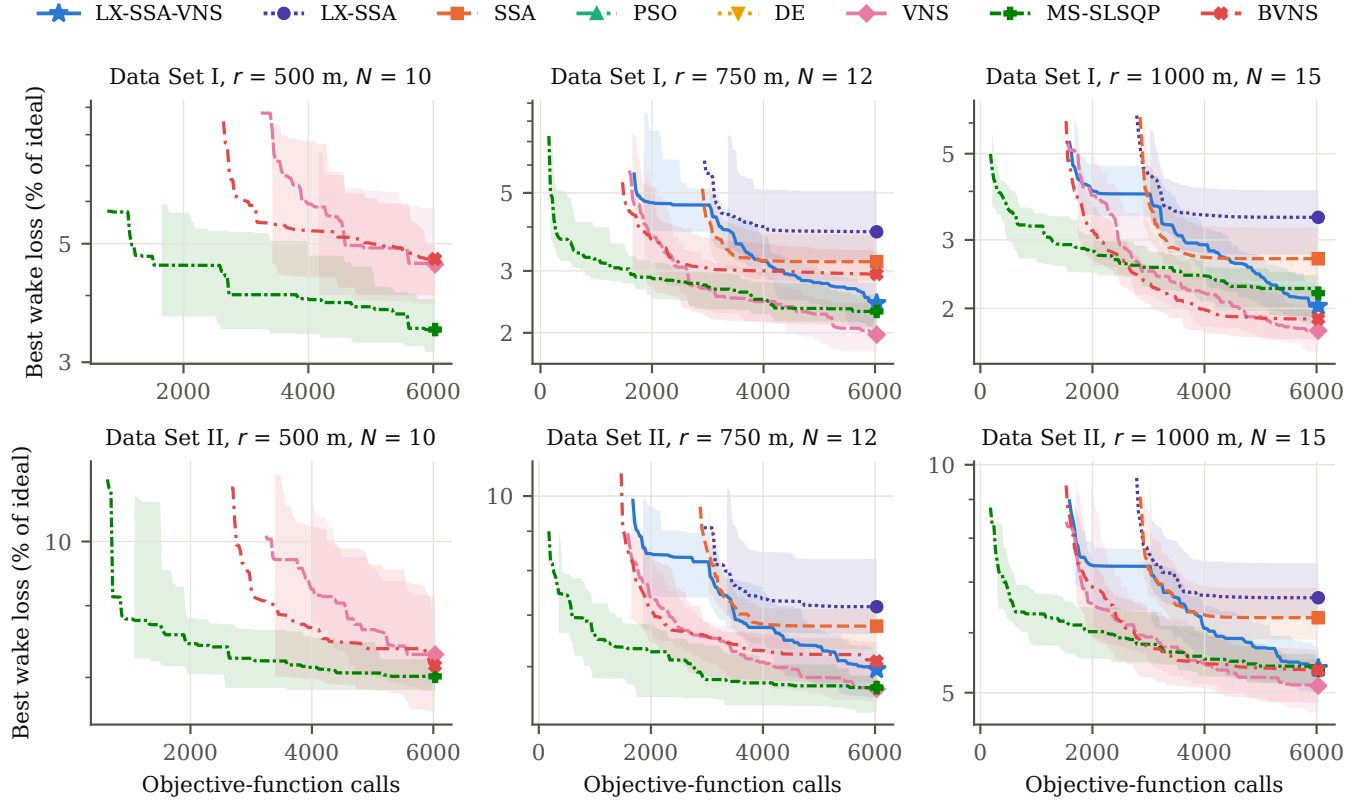


Fig. 8: Convergence for the largest turbine count of each farm (as Fig. 7). Methods for which fewer than half of the runs find a feasible layout have no curve.

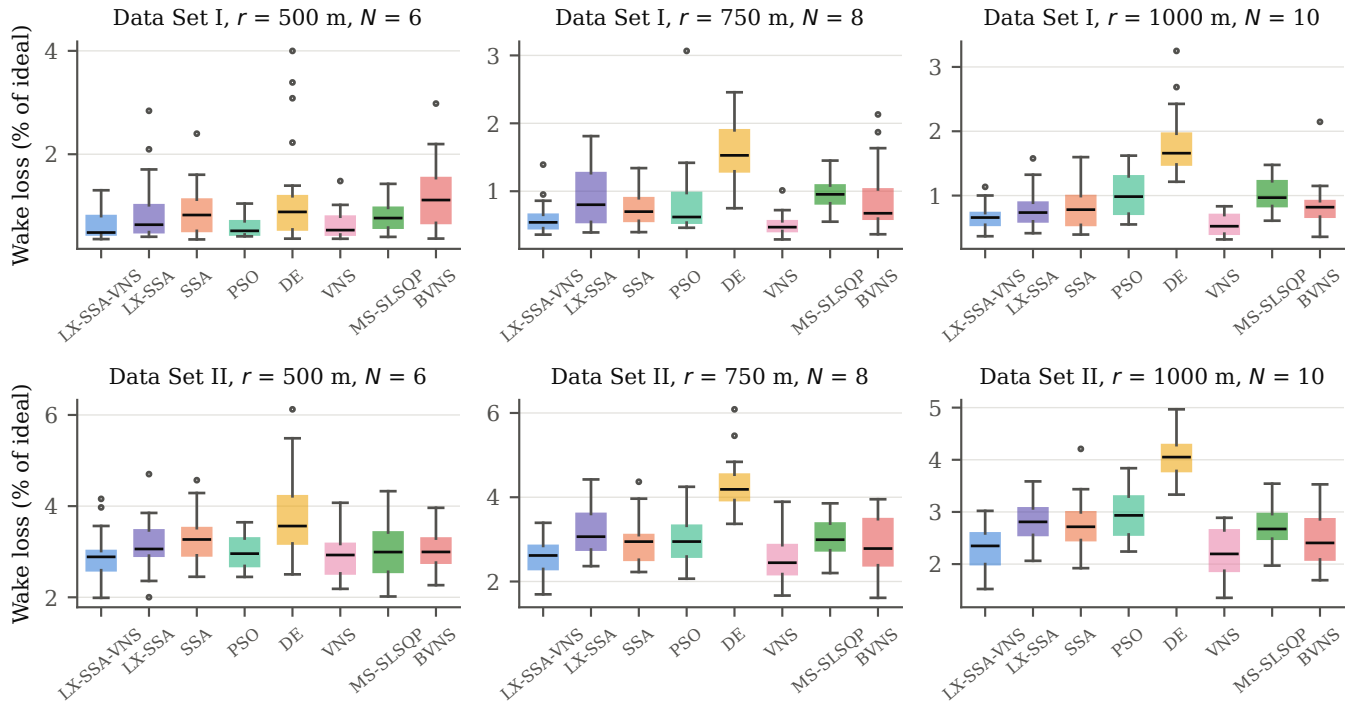


Fig. 9: Distribution of the final wake loss of the feasible runs for moderate turbine counts (30 runs per method).

(653.9 GWh/yr) remains 1.6% below the installed layout. Realistic farm sizes therefore require larger budgets, feasible

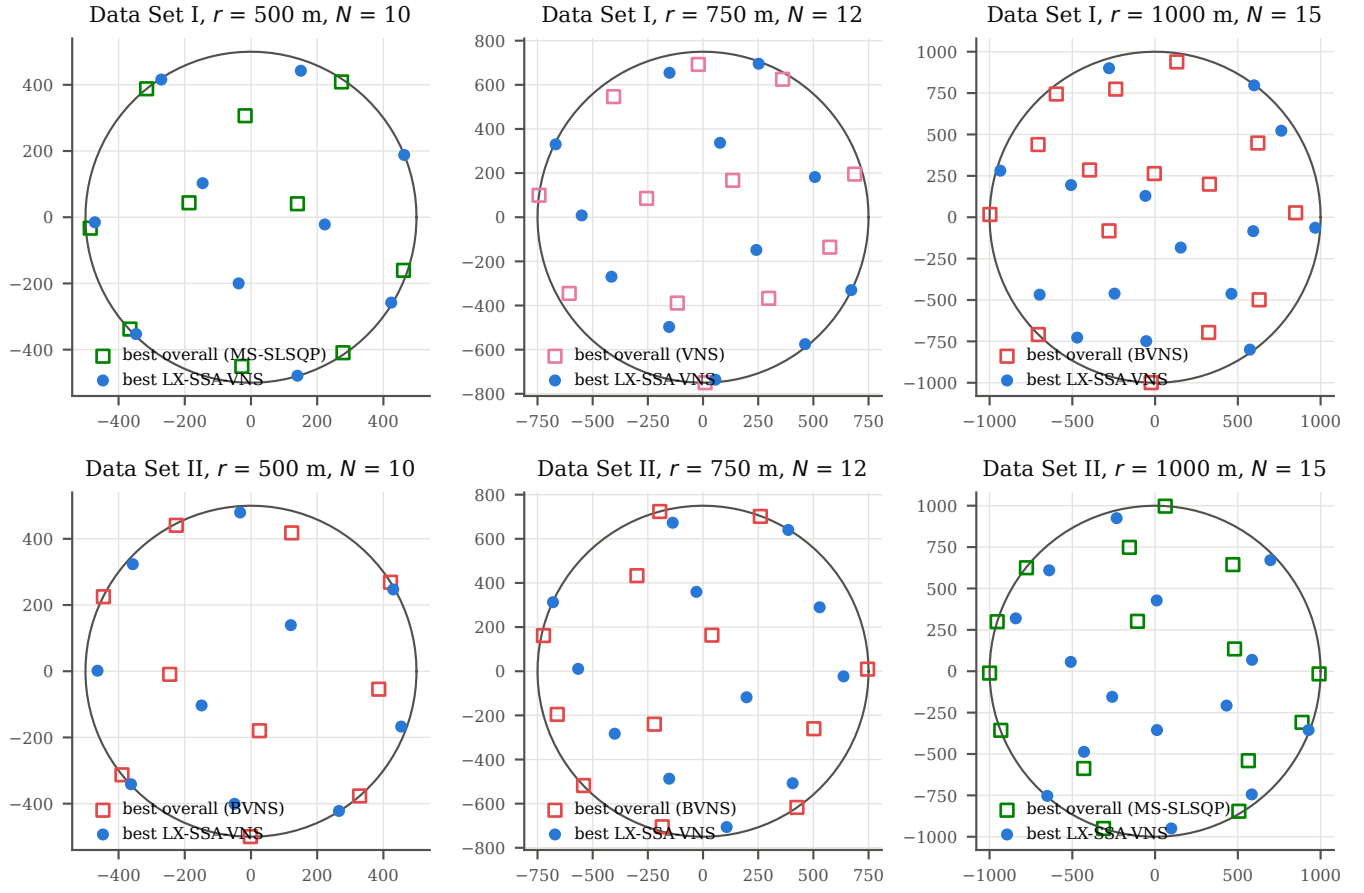


Fig. 10: Best layouts for the largest turbine count of each farm: best LX-SSA-VNS run (filled markers) and best run of any method (open squares, method in the legend). Circle: farm boundary. Coordinates are listed in Appendix A.

TABLE XII: Horns Rev 1 site case (real Vestas V80 power and thrust curves, measured 12-sector wind climate, installed farm outline; Jensen wake with $k = 0.04$; minimum spacing $4D = 320$ m): AEP in GWh/yr, mean (SD) over the feasible runs at 6,030 objective calls (30 seeds for the 16-turbine block, 10 for the 80-turbine farm), mean wake loss relative to the wake-free AEP (“Loss”, %), feasible runs, and Holm-adjusted Wilcoxon signed-rank p of LX-SSA-VNS vs. each method (16-turbine block; run-level Friedman $p = 1.39 \times 10^{-29}$).

Layout / method	16-turbine block (wake-free 149.26 GWh/yr)				80-turbine farm (wake-free 746.3 GWh/yr)		
	AEP	Loss	Feas.	p_{Holm}	AEP	Loss	Feas.
Installed layout	139.51	6.53	—	—	664.6	10.95	—
LX-SSA-VNS	137.59 (0.62)	7.82	28/30	—	—	—	0/10
LX-SSA	136.46 (0.90)	8.58	28/30	0.003	—	—	0/10
SSA	136.66 (1.00)	8.44	30/30	0.059	—	—	0/10
PSO	—	—	0/30	1.30×10^{-8}	—	—	0/10
DE	—	—	0/30	1.30×10^{-8}	—	—	0/10
VNS	137.97 (0.80)	7.57	29/30	0.059	—	—	0/10
MS-SLSQP	136.80 (0.77)	8.35	30/30	0.040	653.9 (2.6)	12.39	10/10
BVNS	137.67 (1.02)	7.77	30/30	0.191	—	—	0/10

initialization or explicit constraint handling, and a regular installed layout is a strong reference.

IX. SENSITIVITY TO THE MODELLING ASSUMPTIONS

A. Power Curve and Wake Model

To test how far the comparison depends on the benchmark power curve and wake model, every feasible final layout was

re-evaluated, without re-optimization, with three alternative models:

- a cubic power curve [45], $f_c(s) = P_{\text{rated}}(s^3 - s_{\text{cut-in}}^3)/(s_{\text{rated}}^3 - s_{\text{cut-in}}^3)$ between cut-in and rated speed;
- the same curve with a 25 m/s cut-out;
- the Gaussian wake model of Bastankhah and Porté-

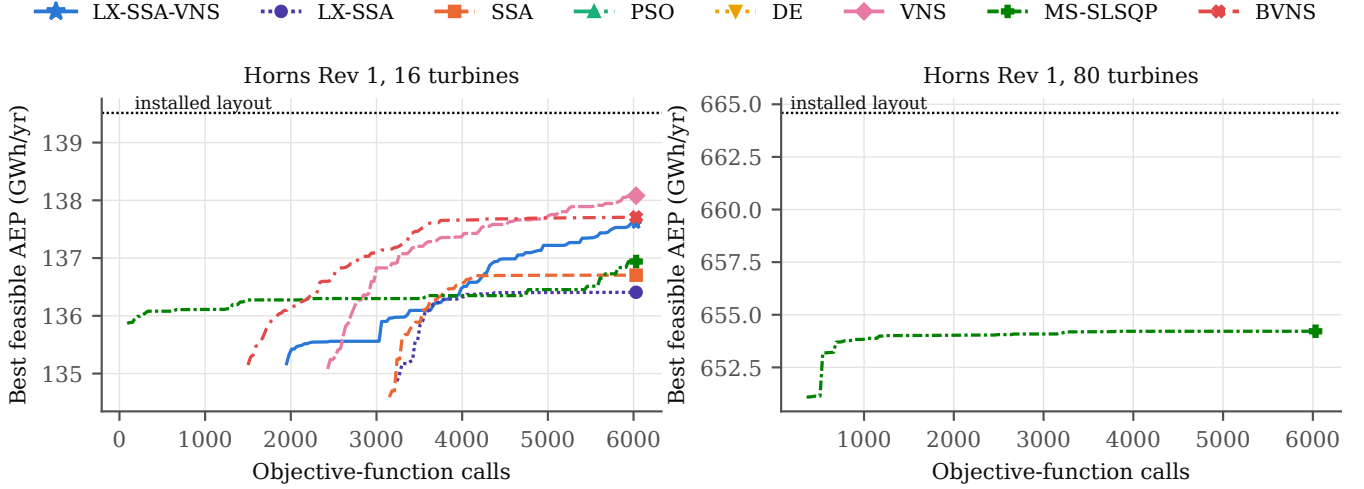


Fig. 11: Horns Rev 1 site case: median best feasible AEP versus objective calls (16-turbine block: 30 runs; 80-turbine farm: 10 runs); dotted line: installed layout. Only MS-SLSQP finds feasible layouts for the 80-turbine farm.

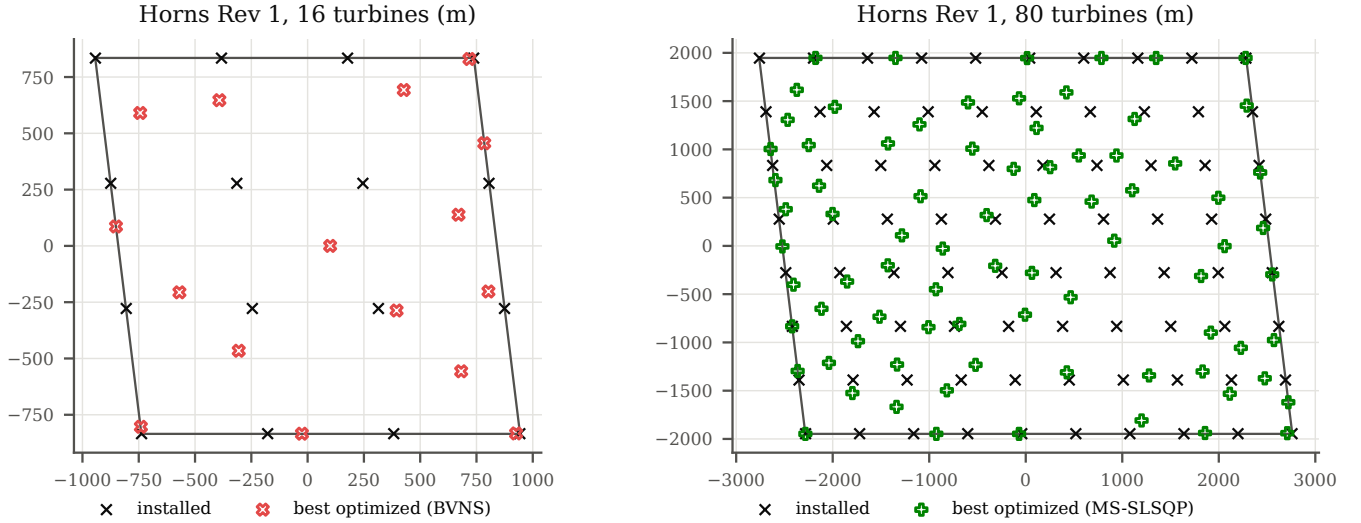


Fig. 12: Horns Rev 1: installed layout (crosses) and best optimized layout (open markers) inside the installed farm outline; local coordinates in metres.

TABLE XIII: Robustness of the results to the benchmark model. All feasible final layouts are re-evaluated (not re-optimized) with alternative power-curve and wake models. Δ : mean relative change of the objective (expected power) with respect to the benchmark model; average rank of each method over the 68 cases (methods with at least five feasible runs); $\bar{\tau}$: mean Kendall rank correlation between the benchmark and alternative orderings of the methods within a case; “Same best”: percentage of cases in which the best method is unchanged.

Model	Δ (%)		Average rank								$\bar{\tau}$	Same best (%)
	DS I	DS II	LX-SSA-VNS	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP	BVNS		
Benchmark (linear curve, Jensen)	+0.0	+0.0	2.72	5.62	5.18	4.79	6.77	2.07	5.28	3.57	—	—
Cubic power curve	-17.7	-36.9	2.78	5.56	5.13	4.68	6.71	2.13	5.37	3.64	0.96	91
Cubic curve with 25 m/s cut-out	-21.5	-37.1	2.79	5.56	5.15	4.65	6.73	2.16	5.34	3.62	0.96	90
Gaussian wake ($k^* = 0.04$)	-0.2	-0.4	3.25	5.75	4.96	3.81	6.72	2.75	4.15	4.62	0.57	46

Agel [46],

$$\delta_{ij} = \left(1 - \sqrt{1 - C_T/(8\sigma^2/D^2)}\right) \exp(-r_{\perp}^2/(2\sigma^2)),$$

$$\sigma/D = k^* x/D + 0.2\sqrt{\beta},$$

with $\beta = \frac{1}{2}(1 + \sqrt{1 - C_T})/\sqrt{1 - C_T}$ and $k^* = 0.04$, evaluated at the rotor center with the same superposition and Weibull scaling as the benchmark.

Table XIII shows two kinds of sensitivity.

TABLE XIV: Largest turbine count for which a layout satisfying the circular boundary and the minimum spacing was constructed (multi-start packing search; a constructive lower bound on geometric capacity), compared with the largest tested N .

r (m)	4D (308 m)	5D (385 m)	6D (462 m)	Largest tested N
500	13	8	7	10
750	26	19	13	12
1000	42	29	21	15

- *Absolute energy is strongly model-dependent.* The linear benchmark curve overstates expected power relative to the cubic curve by 17.7% for Data Set I and 36.9% for Data Set II (21.5% and 37.1% with the cut-out). The absolute energy values of this paper are therefore benchmark values, not turbine-specific AEP predictions.
- *The algorithm comparison is robust to the power curve but less so to the wake model.* Under the cubic curve the within-case orderings are almost unchanged (mean Kendall $\bar{\tau} = 0.96$), and VNS (2.13), LX-SSA-VNS (2.78) and BVNS (3.64) remain the three best methods. Under the Gaussian model the orderings change more ($\bar{\tau} = 0.57$; same best method in 46% of the cases), because layouts optimized for the sharp Jensen wake cone are penalized differently by the smooth Gaussian deficit. VNS (2.75) and LX-SSA-VNS (3.25) nevertheless keep the two best average ranks, whereas BVNS falls to 4.62, behind PSO (3.81) and MS-SLSQP (4.15).

B. Minimum Spacing

The 4D spacing is a legacy benchmark value. Its influence was examined in three ways.

Geometric capacity. Table XIV gives the largest turbine count for which a multi-start packing search constructed a feasible layout; these are constructive lower bounds, and the 500-m values coincide with the known optimal circle-packing results. At 5D the 500-m farm holds at most 8 turbines and at 6D only 7, so the tested 500-m cases with $N \geq 9$ (at 5D) or $N \geq 8$ (at 6D) would be infeasible.

Final layouts. The optimized layouts are frequently active at the 4D constraint. Of the 11,247 feasible final layouts of LX-SSA, SSA, PSO, DE, VNS and MS-SLSQP, 48.7% also satisfy 5D and 35.7% satisfy 6D. For $N = 6$ –10 these fractions fall to 29.6% and 11.3%, and for $N = 11$ –15 to 3.1% and 0%. Optimized layouts therefore cannot be transferred to a site with larger spacing requirements without re-optimization.

Re-optimization. Six representative cases were re-optimized with LX-SSA and SSA at 5D and 6D (Table XV). In the 750-m eight-turbine cases the mean LX-SSA objective falls from 4D to 6D by 0.95% (Data Set I) and 0.22% (Data Set II), and the wake loss rises from 0.88% to 1.85% and from 3.18% to 3.47%. The spacing rule does not reverse the LX-SSA–SSA comparison.

C. Boundary Handling

All methods clip coordinates to the bounding square and penalize points outside the circle (Section III-C). An otherwise identical SSA that instead projects every turbine radially onto the circle attains higher mean objectives than the original SSA in six representative cases at 3,030 calls, by 26 to 649 objective units (Mann–Whitney $p < 10^{-4}$ in every case). The boundary-handling rule therefore affects results noticeably and is held fixed for all methods.

X. COMPUTATIONAL COMPLEXITY

Let N_t be the number of turbines, N_θ the number of wind-direction bins and N_s the number of wind-speed bins. A direct wake evaluation contains the pairwise turbine interactions for every direction and speed bin, so one objective evaluation costs

$$C_{\text{obj}} = \mathcal{O}(N_t^2 N_\theta N_s),$$

and a run with a budget of B objective calls costs $\mathcal{O}(B N_t^2 N_\theta N_s)$ for every method considered here, including both phases of LX-SSA-VNS. Box repair is $\mathcal{O}(N_t)$ per candidate and the spacing check $\mathcal{O}(N_t^2)$; the Laplace draw of LX-SSA and the shaking and local-search moves of VNS are $\mathcal{O}(N_t)$. All of these are dominated by the wake evaluation, which is consistent with the almost identical measured cost per call (Section VIII-I).

XI. SCOPE, LIMITATIONS, AND VALIDITY OF THE EVIDENCE

The results should be interpreted within the following boundaries:

- **Relative performance of the hybrid:** LX-SSA-VNS improves substantially on LX-SSA, SSA, PSO, DE and MS-SLSQP, but it does not outperform the modified VNS at the tested budget, and at the case level its advantage over the original BVNS is not significant. The ablation indicates that its advantage over LX-SSA comes from the VNS phase.
- **Budget and farm size:** all methods were compared at 6,030 objective calls; larger budgets could change the ordering, in particular the value of the exploration phase. For an 80-turbine farm the penalty-based methods, including the hybrid, did not find feasible layouts at this budget.
- **Algorithm settings:** the LX-SSA parameters ($\phi = 0$, $\chi = 1$), the VNS and BVNS settings and the split $\rho = 0.5$ are fixed defaults and were not tuned. Pseudo-gradient and surrogate-assisted methods are discussed but not benchmarked.
- **Wake model and power curve:** Jensen’s model and the piecewise-linear power curve are benchmark choices. Absolute energy values depend strongly on them, and the within-case orderings change under a Gaussian wake model (Section IX-A); layouts were not re-optimized with the alternative models.
- **Spacing and boundary handling:** 4D is a legacy constraint that governs the feasible turbine density, and the boundary-handling rule also affects absolute results.

TABLE XV: Minimum-spacing sensitivity with the authors' LX-SSA and SSA code (6,030 calls, 30 seed-paired runs per cell): mean benchmark objective, Wilcoxon signed-rank p for LX-SSA vs. SSA, and the mean LX-SSA wake loss as a percentage of the ideal objective at 4D and 6D. A superscript gives the number of feasible runs when fewer than 30.

DS	Radius	N	4D (308 m)			5D (385 m)			6D (462 m)			LX-SSA loss (%)
			LX-SSA	SSA	p	LX-SSA	SSA	p	LX-SSA	SSA	p	
1	500	4	56078.8	56086.8	0.237	56082.4	56085.2	0.700	56075.5	56083.9	0.839	0.19 → 0.19
1	750	8	111374.0	111506.5	0.245	111154.1	111163.4	0.871	110314.4 ²⁸	110847.5	0.021	0.88 → 1.85
1	1000	8	111924.6	111947.1	0.887	111953.5	111946.4	0.777	111781.5	111964.7	0.005	0.39 → 0.52
2	500	4	29004.2	29028.7	0.146	28999.5	29019.6	0.229	28981.8	28971.1	0.612	0.88 → 0.96
2	750	8	56659.2	56806.2	0.061	56621.0	56673.3	0.584	56533.9 ²⁸	56690.6	0.038	3.18 → 3.47
2	1000	8	57645.1	57739.3	0.084	57580.7	57655.7	0.140	57475.5	57587.2	0.114	1.50 → 1.79

- **Scope:** N is prescribed in every run. Terrain, heterogeneous turbines, electrical collector systems, grid, acoustic and environmental constraints, and economic objectives are not modeled.

XII. CONCLUSION

This paper proposes LX-SSA-VNS, a two-phase hybrid in which the previously published Laplacian Salp Swarm Algorithm (LX-SSA) [30] explores the continuous layout space and variable neighbourhood search then intensifies the best layout found. The hybrid is applied to the continuous WFLOP with a Jensen–Weibull benchmark model and explicit constraint handling. It is evaluated against LX-SSA, SSA, PSO, DE, the original basic VNS (BVNS), a modified VNS and a gradient-based multistart SLSQP solver on 68 benchmark cases (16,320 seed-paired runs at equal budgets) and on the Horns Rev 1 site. The results support five conclusions.

- *Hybridization substantially improves LX-SSA.* LX-SSA-VNS is significantly better than LX-SSA in 35 of 68 cases and never worse. It lifts LX-SSA from the seventh to the second average rank among eight methods (5.68 to 2.74) and achieves a feasibility rate of 98.1%.
- *The hybrid outperforms the other population-based and gradient-based references.* It is significantly better than SSA, PSO, DE and MS-SLSQP in most cases, and at the case level it ranks significantly better than all of them.
- *The modified VNS remains the best method at this budget.* The hybrid is statistically indistinguishable from it in 53 cases but worse in 15 dense cases. The ablation shows that the gain comes from the intensification phase, and that assigning more budget to LX-SSA is harmful.
- *The original basic VNS is a strong baseline.* BVNS ranks third (3.62), is feasible in every run, and is better than the hybrid in seven cases (small or very dense layouts) and worse in 12. The modified VNS beats it in 27 cases with $N \geq 6$, while BVNS is better in eight cases with $N \leq 4$, so the cheaper turbine-wise search pays off for larger layouts.
- *A similar ordering holds at a real site.* In a 16-turbine block of Horns Rev 1 the hybrid is better than LX-SSA and MS-SLSQP and statistically indistinguishable from SSA, VNS and BVNS. No method improves on the installed layout within the budget, and for the full 80-turbine farm only the constraint-aware MS-SLSQP finds feasible layouts.

- *The comparison depends on the wake model more than on the power curve.* The algorithm ordering is insensitive to the power curve but changes considerably under a Gaussian wake model, and absolute energy values depend strongly on both.

Future work should use a stagnation-triggered switch between the two phases, feasibility-preserving initialization for dense farms, larger budgets for realistic farm sizes, and re-optimization under higher-fidelity wake and power models.

APPENDIX A

COORDINATES OF THE BEST LAYOUTS

Table XVI lists the turbine coordinates (m, farm center at the origin) of the best layout found by any method for the largest turbine count of each farm (Fig. 10). The final coordinates and convergence curves of all 16,320 runs and the Horns Rev 1 layouts are provided as machine-readable files with the revision.

Compliance with Ethical Standards

Conflict of Interest: The authors declare that they have no conflict of interest.

Ethical Approval: This article does not contain any studies with human participants or animals performed by any of the authors.

Informed Consent: Not applicable.

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TABLE XVI: Coordinates of the best layout for the largest turbine count of each farm (method and benchmark objective given).

DS	r (m)	N	Method	Objective	Coordinates (ρ_i, σ_i) (m)
1	500	10	MS-SLSQP	138171.2	(-315.2, 388.1); (277.9, -409.2); (-187.2, 43.6); (-27.6, -450.5) (-17.7, 306.6); (460.7, -160.4); (-365.7, -337.8); (-485.5, -33.0) (273.5, 408.8); (140.3, 40.7)
1	750	12	VNS	166626.2	(9.5, -749.9); (-255.6, 85.4); (574.3, -135.8); (134.3, 167.2) (-606.4, -345.2); (-116.6, -388.9); (297.6, -367.4); (687.2, 195.4) (-21.4, 692.3); (-743.2, 99.5); (-405.1, 546.3); (360.5, 625.1)
1	1000	15	BVNS	208307.4	(849.2, 26.7); (-396.0, 285.5); (-707.1, 439.0); (-278.5, -82.2) (-237.9, 773.9); (-596.9, 744.5); (323.7, -696.2); (-705.5, -708.7) (131.1, 938.4); (628.6, -498.9); (-23.6, -999.4); (620.8, 448.3) (328.2, 199.8); (-997.9, 16.7); (-4.6, 263.8)
2	500	10	BVNS	68269.3	(-445.8, 225.2); (25.6, -179.7); (421.4, 268.9); (-225.6, 440.8) (123.1, 417.9); (386.0, -54.2); (328.2, -376.7); (-245.5, -9.4) (-389.7, -312.9); (-1.9, -499.6)
2	750	12	BVNS	83609.6	(-220.7, -240.0); (-184.6, -704.7); (425.4, -617.0); (-196.1, 723.8) (745.5, 9.5); (502.1, -260.4); (259.7, 701.5); (40.1, 163.4) (-660.6, -195.5); (-541.5, -518.0); (-299.6, 433.0); (-722.9, 161.4)
2	1000	15	MS-SLSQP	105183.7	(480.1, 134.7); (887.9, -308.8); (-932.2, -356.9); (-954.1, 299.4) (-999.9, -11.2); (469.8, 644.1); (-777.1, 625.3); (-106.5, 301.6) (-155.3, 748.4); (-430.9, -587.6); (-311.4, -949.7); (506.2, -845.8) (61.2, 997.2); (990.7, -16.0); (563.2, -540.5)

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